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A multi-criteria analytics approach for circular supplier selection in sustainable and human-centric supply chains

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ABSTRACT

This study integrates circular economy principles and Industry 5.0 dimensions into the supplier selection problem, introducing the concept of "Circular Supplier 5.0" selection within an analytics-driven supply chain context. A novel multi-criteria decision-making approach is developed by integrating the general best–worst method (GBWM), interval numbers, and a Supermatrix structure. Six main factors (resiliency, human-centricity, economic, circular, social, and Industry 4.0), along with 27 sub-factors, are proposed to evaluate Circular Supplier 5.0 performance in the construction industry. In the proposed framework, interval GBWM is used to determine the independent weights of factors and sub-factors, capture interdependencies among factors, and assess the relative performance of suppliers across sub-factors. A Supermatrix-based structure is then employed to compute the final dependent weights and overall supplier rankings. To demonstrate applicability, six suppliers from a Malaysian construction company are evaluated. The results show that, compared with traditional approaches that overlook Industry 5.0 and circularity considerations, the proposed model highlights the importance of incorporating these dimensions into supplier evaluation. The findings indicate that security, cost, quality, capability, information sharing, and traceability are the most influential sub-factors. Expert validation further supports the robustness of the results and confirms the practical relevance of the proposed framework for real-world supply chain analytics and supplier selection decisions.

1. Introduction

One of the influential factors in gross domestic product (GDP) is the construction industry [1]. This industry invests significantly in civil, transportation, energy, and buildings infrastructures and, as one of the dynamic industries, it leads to the progress and development of societies [2]. In recent years, for instance, researchers have paid attention to supply chain management in the construction industry [3]. Although the construction industry leads to economic prosperity, GDP growth, and positive social transformations, it may result in many biological and environmental damages. The construction industry is one of the largest consumers of natural resources in the world and is considered a serious threat to the planet [4]. Now, the question is how to reduce or eliminate the harmful environmental effects of this industry. Implementing sustainable practices in this industry is a practical solution to overcome this challenge [5,6]. For example, in order to apply sustainable practices in the construction industry, many developed and developing countries have established strict environmental regulations in this industry and have had the industry owners comply with these regulations [7,8]. Furthermore, the fifth industrial revolution (Industry 5.0) principles have been recently integrated into construction project management, driving research attention to human-centered and resilient supply chains [9,10].

In addition, the construction industry has always faced challenges related to quality, cost, and time. Now, another question is raised: Do sustainability practices have an effect on reducing costs, increasing quality, and reducing project implementation time in this industry? Sustainability practices and specifically circular economy (CE) practices

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can address environmental and economic (i.e., quality, cost, and time) concerns [11]. Research on CE applications in construction supply chains, nonetheless, has shown gains in resource recovery and waste reduction, underscoring the need for the application of advanced decision-making methods to assess supplier performance [12,13].

One of the strategic measures to increase sustainability in various industries is cooperation with sustainable suppliers [14]. Suppliers are among the upstream players of the supply chain and they, as the core of sustainability, have a significant impact on the economic, social, and environmental performance of industries. Cooperation with sustainable suppliers leads supply chain activities towards sustainable practices and is in line with the principles of Industry 5.0. Dwivedi et al., [15]. Industry 5.0 is a new concept based on the three principles of human-centricity, sustainability, and resilience, which aims to combine Industry 4.0 technologies with the innovative thinking of experts [16,17]. The concept of sustainability in Industry 5.0 is based on three economic, social, and environmental pillars, and aims to create a balance between cost minimization (profit maximization), the minimization of negative environmental effects, and the maximization of positive social effects in different industries [18]. Industries are always exposed to disruptions caused by natural disasters, economic fluctuations, technological changes, etc. Resilience, as one of the principles of Industry 5.0, is a criterion, through which one can evaluate the ability and flexibility of industries in facing these disruptions. In other words, a resilient system is a flexible one that quickly adapts to unforeseen challenges and changing market demands [19]. The Human-centricity dimension of Industry 5.0 focuses on creativity, expertise, and well-being of human, and considers the workforce as the heart of industrial processes [20]. Industry 5.0 ties the technologies based on Industry 4.0 with human expertise and creates innovative productive environments by preserving human values [21].

Therefore, cooperation with the suppliers who value CE principles and Industry 5.0 criteria in their daily activities not only paves the way for organizations to achieve sustainability goals, but also improves resource efficiency, waste management, and conditions for the application of advanced human-centered technologies. Circular supplier 5.0 selection in the construction industry leads to an increase in the flexibility, efficiency of resources, quality of projects, and speed of project execution and a reduction in carbon footprint, costs, and waste.

Now it is time to respond to the question of how to select the best circular supplier 5.0 in the construction industry. The circular supplier 5.0 selection problem is a specific type of supplier selection problem that uses criteria based on CE principles and Industry 5.0 in the supplier evaluation process. To achieve the best circular supplier 5.0, (1) appropriate criteria should be identified and (2) an efficient approach for weighting these criteria and supplier prioritization should be developed. Therefore, in this article, we intend to advance the supplier selection problem from two conceptual and methodological perspectives. The literature review shows that, so far, many articles have been presented in the field of supplier selection problem, many of which such as Koc et al. [22], Sarkar et al. [23], and Singh et al. [24] have considered sustainability criteria, some have focused on circular criteria [25,26], and a few have considered Industry 5.0 criteria [27,28]. From a conceptual perspective, to the best of our knowledge, only two articles, namely Nayeri et al. [29] and Nayeri and Sazvar [30] have considered the criteria of CE and Industry 5.0 in the supplier selection problem. In this vein, for the first time, this article considers factors of resiliency, human-centricity, economic, circular, social, and Industry 4.0 for the supplier selection problem in the construction industry, utilizing CE principles and Industry 5.0 dimensions. In addition, methodologically, for the first time, by integrating the general best-worst method (GBWM), interval numbers, and the Supermatrix structure, a new approach is presented for weighting the factors and prioritizing the alternatives in intertwined hierarchical networks under uncertainty. The literature review shows that researchers have used a combination of weighting methods and approaches such as the decision making trial and evaluation laboratory (DEMATEL), weighted influence non-linear gauge system (WINGS), etc. to calculate the dependent weights of criteria. However, there are few methods such as GBWM that apply an integrated structure to calculate the dependent weights of criteria [31]. Although GBWM has solved the problem of integration in calculating the dependent weights of criteria, it cannot calculate the scores of alternatives in an integrated manner. Govindan et al. [32] have stated that the use of Supermatrix is a suitable solution to calculate the dependent weights of criteria and the scores of alternatives in an integrated manner. The developed approach is effective in both problems of weighting factors and ranking alternatives, and unlike GBWM, it provides an integrated structure for scoring alternatives and ranking them. It should be noted that, in the proposed approach, we have used interval numbers to deal with uncertainty. Indeed, these numbers make pairwise comparisons flexible for experts, are easily integrated with multi-criteria decision-making (MCDM) methods, and have low computational complexity. Finally, the effectiveness of the developed approach is evaluated through its implementation in a construction industry in Southeast Malaysia.

In general, the objectives of this research can be stated as follows:

- Identifying the sub-factors of resiliency, human-centricity, economic, circular, social, and Industry 4.0 to evaluate suppliers from the perspective of CE and Industry 5.0;
- Developing a novel MCDM approach to calculate the dependent weights of criteria and alternative scores integrally by the combination of GBWM, interval numbers, and Supermatrix structure;
- Examining the applicability of the proposed approach through its implementation in a construction company in Malaysia.

The remainder of the article is organized as follows. Section 2 provides the motivation and contributions. Sections 3 and 4 are dedicated to the literature review and the proposed approach, respectively. The case study is provided in Section 5. Section 6 is assigned to the discussion. Theoretical and managerial implications are presented in Sections 7 and 8. Finally, the conclusion is provided in Section 9.

2. Motivation and contributions

Although the application of MCDM methods in supplier selection has dramatically increased, the related literature in this area is not free of gaps, which justifies the conduct of the current approach. It is noteworthy that the interdependencies among criteria has been already extended by GBWM, but GBWM has not been integrated with a Supermatrix structure in any study to go for a simultaneous computation of dependent weights and alternative scores. Existing approaches in this field calculate independent weights, interdependencies between criteria, and supplier scores in separate phases, but in our approach, a unified weighting-prioritization structure will be used for this purpose. In addition, most of the studies in this light have used fuzzy set theory such as interval type−2 fuzzy [33], hesitant fuzzy [34], and so forth to deal with uncertainty. It is true that such fuzzy variants can lead to a final picture of detailed ambiguity; however, substantial computational complexity often arise, a high level of expertise is needed in fuzzy logic, and interpretability challenges show up for those practitioners who lack familiarity with defuzzification procedures and membership functions. Due to the fact that both methodologists and supply chain managers use MCDM methods to a great extent, the practicability and transparency of the decision-making framework assume high importance. Accordingly, interval numbers are knowingly adopted as a more straightforward and accessible alternative. Indeed, the bounded nature of expert uncertainty is maintained by interval numbers, the seamless integration of interval numbers with linear programming formulations is carried out, and the interpretation of these interval numbers becomes an easy task for industry professionals with no need for any formal training in fuzzy logic. Besides, uncertainty propagation is maintained by the current

interval-based framework throughout the whole decision-making process with no premature defuzzification, which creates a pragmatic balance between applicability and rigor. Finally, it should be noted that no analytics-driven approach in prior studies has managed to go for the simultaneous handling of uncertainty, hierarchical interdependencies, and the integrated evaluation of six dimensions, namely human-centricity, social, resiliency, Industry 4.0, circular, and economic. By covering these gaps, the current study aims at developing a novel approach that specifically integrates a Supermatrix structure, GBWM, and interval numbers to present a strong unified uncertainty-resilient decision approach for circular supplier 5.0 selection.

3. Literature review

The literature review section of this article consists of two subsections. The first sub-section explores the supplier selection problem and its evolution over time. The second sub-section focuses on the evaluation criteria of circular supplier 5.0 and introduces a complete set of criteria.

3.1. Evolution of supplier selection problem

The supplier selection problem is a decision-making problem whose goal is to select the best supplier using multiple and conflicting criteria. MCDM methods are among the most widely used methods in this field [35]. In the traditional supplier selection problem, the objective is to select the best supplier using economic criteria such as cost, quality, delivery, etc. With the rise of environmental concerns, a new concept, called "green supplier" selection emerged and researchers considered environmental criteria along with economic criteria to evaluate suppliers [36]. Sahoo and Goswami [37] have provided a comprehensive review of recent articles in the field of green supplier selection using MCDM methods. As the concept of sustainability became widespread, supplier selection underwent serious changes and was significantly deviated from its traditional version. In this vein, researchers incorporated economic, environmental, and social criteria into the supplier evaluation process and defined a new concept, called "sustainable supplier" selection. There is a rich literature on sustainable supplier selection using MCDM methods in various industries [24,38,39].

In recent years, with the emergence of concepts such as CE and Industry 4.0, the supplier selection problem has grown significantly. Kannan et al. [40] pioneered the combination of CE concept into sustainable supplier selection, introducing the novel concept of "circular sustainable supplier" selection. Their work laid the foundation for subsequent studies, with researchers such as Ecer and Torkayesh [41], Echchafj et al. (2023), and Mishra et al. [25] advancing the field by employing MCDM methods to solve circular sustainable supplier selection problems. In this light, some scholars such as Xie et al. [42], Kusi-Sarpong et al. [43], and Ali et al. [44] took a step further and included both CE and Industry 4.0 concepts in the sustainable supplier selection problem.

Although the nascent concept of Industry 5.0 provides new opportunities for developing the supplier selection problem, the literature review shows that the supplier 5.0 selection problem has still received little attention from researchers. Lo [27] is the pioneer of this field. He has developed a decision support system to evaluate the suppliers of a medical equipment manufacturing industry by applying Industry 5.0 criteria and a data-driven decision-making concept. Bayatzadeh and Talaie [45] applied resilience, sustainability, and Industry 5.0 criteria to evaluate suppliers in the steel industry. They used the fuzzy BWM method to weight the criteria and prioritize suppliers. In this vein, Asadi et al. [28] examined the supplier selection and order allocation problem from the perspective of viability and Industry 5.0. They identified evaluation criteria from the perspective of viability and Industry 5.0, and calculated their weights by stochastic fuzzy BWM. They used stochastic fuzzy TOPSIS method to rank suppliers. Similarly, a hybrid approach based on MCDM methods was presented by Mashhadi et al. [46] to evaluate suppliers from the perspective of Industry 5.0 and viability. They used lexicographic stochastic fuzzy BWM to calculate the weights of the criteria and stochastic fuzzy normalized weighted sum method to rank suppliers.

Nayeri et al. [29] and Nayeri and Sazvar [30] have taken it a step further by incorporating criteria of Industry 5.0 and CE into the supplier selection problem. Nayeri et al. [29] evaluated suppliers of an automotive industry in Iran by employing Industry 5.0 and CE criteria. They utilized the stochastic fuzzy BWM to calculate the weights of the criteria and ranked the suppliers using the stochastic fuzzy complex proportional assessment (COPRAS) method. Similarly, Nayeri and Sazvar [30] used Industry 5.0 and CE criteria to evaluate suppliers of a home appliance industry in Iran. They extracted evaluation criteria from the aspects of digitalization, sustainability, resilience, human-centricity, CE, and general, and determined their weights using the stochastic BWM. Finally, suppliers were ranked through the stochastic COPRAS method. As a conclusion, the literature review reveals a research gap in the field of supplier 5.0 selection. Integrating the concepts of CE and Industry 5.0 with the supplier selection problem can help the development of this field. This article presents a comprehensive supplier evaluation framework specifically tailored for the construction industry, integrating five key dimensions: sustainability, CE principles, Industry 4.0 technologies, resilience, and human-centricity. While Nayeri et al. [29] and Nayeri and Sazvar [30] have explored similar integrations of CE and Industry 5.0 concepts, our work distinguishes itself through its industry-specific criteria selection and application to construction supply chains. Furthermore, a new MCDM approach is presented to prioritize circular supplier 5.0 in the construction industry.

From a methodological perspective, a review of the literature shows that although the factors involved in the supplier selection problem are inherently intertwined, some papers consider them independently to reduce computational complexity. On the other hand, most papers use a combination of two or more methods to evaluate and rank suppliers. In this vein, for the first time, a unified weighting-prioritization approach based on fuzzy GBWM for supplier selection problem was developed by Tavana et al. [47]. Their approach considered interdependencies between factors in the weighting and prioritization process. In this regard, this article develops a new unified weighting-prioritization approach by integrating GBWM, interval numbers, and the Supermatrix structure. Table 1 provides the similarities and differences between our article and the existing articles in the literature from the perspective of methodology and evaluation criteria.

3.2. Circular supplier 5.0 selection criteria

In this subsection, by reviewing the literature, six factors, including resiliency, human-centricity, economic, circular, social, and Industry 4.0, along with 27 sub-factors are introduced, which are given below.

3.2.1. Resiliency

- Backup suppliers

One of the strategies for increasing the resilience of supply chains is cooperation with backup suppliers [48]. In the event of a disruption in the supply chain, the company is supported by these suppliers. It should be noted that these suppliers are not the main suppliers of the company and are considered as a contingency alternative to increase resilience and reduce supply chain risks [49]. It is noteworthy that usually the main suppliers also identify a set of potential suppliers to increase their resilience and are supported by them when needed. Therefore, in order to increase its resilience, a supplier must interact with several backup suppliers operating in its field.

Table 1
Supplier selection articles incorporating Unified weighting-prioritization methods or Industry 5.0 criteria.

Author(s)	Approach/method	Supplier 5.0 selection	Uncertainty			Unified weighting-prioritization	Interdependencies between factors	Factor						Construction industry
			Interval/ grey	Fuzzy	Stochastic			Resiliency	Human-centricity	Economic	Social	Circular	Industry 4.0	
Lo [27]	Criterion importance through intercriteria correlation method and Modified classifiable TOPSIS	√	-	-	-	-	√	-	-	√	√	-	-	-
Bayatzadeh and Talaie [45]	Fuzzy BWM	√	-	√	-	-	-	√	-	√	√	-	-	-
Tavana et al. [47]	Fuzzy GBWM	-	-	√	-	√	√	-	-	-	-	-	-	-
Asadi et al. [28]	Stochastic fuzzy BWM and stochastic fuzzy TOPSIS	√	-	√	√	-	-	√	√	√	√	-	√	-
Mashhadi et al. [46]	Lexicographic stochastic fuzzy BWM and stochastic fuzzy normalized weighted sum	√	-	√	√	-	-	√	√	√	√	-	√	-
Nayeri and Sazvar [30]	Stochastic BWM and stochastic COPRAS	√	-	-	√	-	-	√	√	√	√	√	√	-
Nayeri et al. [29]	Stochastic fuzzy BWM and stochastic fuzzy COPRAS	√	-	√	√	-	-	√	√	√	√	√	√	-
This article	Suppermatrix-based interval GBWM	√	√	-	-	√	√	√	√	√	√	√	√	√

- **Flexibility and agility**
  One of the indicators of a resilient responsive supplier is flexibility and agility. Flexibility and agility refer to the ability of suppliers to meet the changing demands of customers as quickly as possible, which represent variation in the quantity or type of order [50,51].
- **Multiple sources**
  One of the resilience and risk reduction strategies in the supply chain is multiple sources [52]. Suppliers who are single-source and depend only on one source for their supplies, are always exposed to the risks of price increases, supply disruptions, quality reduction, and so on. Experienced suppliers use the multiple sources strategy to increase their resilience and competitive power and do not risk their interests by relying on a single source.
- **Data-driven analytical methods**

Today, suppliers do not use traditional methods to identify risks and analyze them. Data-driven algorithms and methods enable suppliers to identify risks with deep and comprehensive analysis and reduce their vulnerability by focusing on them [53,54].

3.2.2. *Human-centricity*

- Workforce health and safety
  Paying attention to the health and safety of employees is one of the social and human-centricity criteria in the supplier selection problem, which is at play in almost all sustainable supplier selection articles [55]. This criterion requires suppliers to pay attention to health regulations and provide an accident-safe work environment for their workforce.
- **Ethical labor practices**
  Paying attention to ethical labor practices is one of the social and human-centricity criteria in the field of sustainable supplier selection, which some articles like Yassine and Singh [56] and Barbanti et al. [57] have used. This criterion requires suppliers to avoid unethical labor practices such as forced and child labor.
- **Human rights**
  One of the human-centricity criteria in supplier selection problem is human rights, which has been examined in most articles in this field. This criterion protects the rights and interests of the workforce and stakeholders, and requires suppliers to provide fair working conditions, pay fair wages, adhere to labor laws, and so forth [58].
- **Employee training**

Employee training is one of the human-centricity criteria in the field of supplier evaluation. The suppliers who focus on training their employees and allocate part of their organizational budget to this item would help to increase the productivity of their organization and improve the quality of their services and products [59]. In fact, the trained workforce performs its duties more efficiently by applying its skills, and this leads to an increase in productivity. On the other hand, the probability of error occurrence by the trained personnel is low and this ensures the achievement of higher quality products or services.

3.2.3. *Economic*

- Cost
  One of the most key criteria for evaluating suppliers is cost or price, which has been used in almost all articles in the field of supplier selection. About 70% of the final price of products is related to the price of its raw materials [60]. This means that the supplier plays a major role in the price of the final products. Therefore, cooperation with the suppliers who use mechanisms to reduce the costs of their services or products is important and increases their competitive power [61].
- **Delivery**
  One of the key criteria for evaluating suppliers, especially in the field of construction, is the delivery criterion [62]. This criterion shows how capable a supplier acts in delivering products or services within a specified time window [63]. A supplier that neglects delivery time, delivery rate, and delivery reliability will disrupt the operations of the buyers and impose huge costs on them.
- **Quality**
  Another key economic criterion for evaluating suppliers is the quality criterion, which has been employed in almost all articles in the field of supplier selection [64]. This criterion reveals how well the products or services provided by a supplier conform to predetermined standards. This criterion directly affects customer satisfaction [65]. Suppliers can increase the quality of their products and services by applying strategies such as training employees, improving and optimizing processes, customer feedback, and using advanced technologies.
- **Capability**
  The capability criterion in the supplier selection problem refers to the technological and financial ability of suppliers [66]. In general, the financial capability of suppliers means their ability to manage financial resources, manage financial risks, and invest in required infrastructure. Technological capability refers to the technical ability, infrastructure, expertise, and capacity of suppliers. The use of modern and advanced machines has a significant effect on the technological performance of suppliers, which requires access to financial resources.
- **Reputation**

A supplier's reputation is built based on its performance in previous projects. Each supplier earns a credit and position in the market by cooperating with different buyers [67]. Obtaining the satisfaction of customers, employees and stakeholders, the rate of successful projects, adherence to commitments, etc. are among the factors that measure suppliers' reputation. The financial and technological capabilities of suppliers can have a significant impact on the reputation of suppliers.

3.2.4. *Social*

- Information sharing
  Suppliers can gain the trust of supply chain stakeholders by increasing transparency in their processes. One of the ways to create transparency in the supply chain is to share information with stakeholders (Lo, 2024). Information sharing leads to strengthening the relationship between the supplier and the buyer, increases the level of responsiveness, and helps the parties make enlightened decisions [68].
- **Diversity and inclusion**
  One of the social criteria in the supplier evaluation process is diversity and inclusion, which reflects the supplier's social responsibility and commitment [69]. The meaning of diversity and inclusion in the supplier selection problem is to measure the supplier's ability to provide a fair and non-discriminatory environment for its employees regardless of their race, gender, sexual orientation, and age.
- **Employment practices**
  Another important social criterion in the supplier selection problem is the employment practices criterion, which evaluates the supplier's performance in hiring and managing the workforce [70]. This criterion requires suppliers to hire efficient employees for each job position and to create a fair working environment for all employees by adhering to labor laws.
- **Job creation**

Low unemployment rate is one of the signs of developed societies [71]. The suppliers who pay attention to job creation actually contribute to the development of communities. The criterion of job creation in the supplier selection problem means the supplier's ability to create jobs,

especially for local people, promote their businesses, and generally develop the workforce [72].

3.2.5. Circular

- Energy consumption
  In a supply chain, a significant portion of energy is consumed at the supplier echelon. The construction industry is among the industries that are highly dependent on energy [73]. This criterion measures the supplier's performance in the consumption of water, electricity, fossil fuels, etc. A supplier using mechanisms for energy management, in addition to reducing its costs, brings less negative environmental effects (Lo, 2024; [74]).
- **Carbon emission**
  One of the most key criteria in the green supplier selection problem is the carbon emission criterion. This criterion measures the amount of carbon emitted due to supplier activities. The construction industry is one of the industries with the highest level of carbon emissions [75]. A significant part of the carbon in this industry is related to activities such as the production of construction materials, recycling of construction waste, and transportation [76,77].
- **Clean technologies**
  The use of old and low-efficiency technology is one of the main factors contributing to carbon emissions. In addition to protecting the environment, the use of advanced eco-friendly technologies leads to cost reduction in the long term [78].
- **Waste management**
  The purpose of waste management criterion in the supplier selection problem is to evaluate the performance of suppliers in the management of produced waste. For this purpose, the supplier's performance is measured by factors such as using practices to minimize waste production, reduce waste disposal, and enhance waste recycling [66].
- **Green research and development (R&D)**
  The R&D unit in any organization is considered as the central core of that organization, which provides innovative and creative ideas. Green R&D in the supplier selection problem means evaluating the performance of the supplier's R&D unit from an environmental perspective. In this regard, it is determined how much a supplier's R&D activities have been effective in producing environmentally friendly products, reducing carbon emissions, and generally protecting the environment [79].
- **Eco-friendly packaging**

Almost all the products produced by the suppliers are delivered to the buyers in a packaged format. The packaging process is an important process in all industries as it significantly affects the environment. The use of recyclable materials for packaging, applying eco-friendly packaging technologies, and the reuse of waste resulting from the packaging process are among the ways to achieve an eco-friendly packaging [13].

It is noteworthy that CE consists of a group of environmental sustainability practices and some commonalities exist between CE criteria and general environmental dimensions. However, there are significant differences between these two concepts, especially with regard to their focus and objectives. The main focus of CE is on the maintenance of materials and resources value during the process of production, consumption, loops recycling, waste reduction, and promotion of circular business models. On the contrary, the environmental dimension refers to any influence that may be exerted on the environment by industrial activities at large. These activities may include soil pollution, air, water, and resource depletion [80,81]. The factors considered in this study for the CE dimension included carbon emissions, energy consumption, eco-friendly packaging, and waste management and have some shared features with some environmental aspects. However, these factors have been considered in such a way that the dynamics of sustainable design, innovation in business models, and material cycles are assigned credit and, thereby, something beyond minimizing negative environmental impacts assumes importance. Just renaming these criteria as "environmental aspects" removes the transformative and circular approach from the focal center, that is Industry 5.0. Moreover, environmental features are considered to be interconnected with CE indicators in this study, which shows their key role of in sustainable industrial transformation. But, the main focus in this study has been deliberately placed on CE since it is regarded as an important driver for digital and sustainable innovation in Industry 5.0 that enhances forward-looking business models, which are totally different from the conventional environmental management approaches. Therefore, in this article, we use the concept of "Circular Supplier 5.0" instead of sustainable/green supplier.

In general, in this article, "circular supplier selection" means considering circular criteria in the supplier selection process. Similarly, "supplier 5.0 selection" refers to a process that incorporates Industry 5.0 criteria into the supplier selection problem. Obviously, "circular supplier 5.0 selection" considers both Industry 5.0 and circular criteria in the supplier selection problem.

3.2.6. Industry 4.0

- **Platforms**
  The application of Industry 4.0 technologies can lead to the improvement of relations between suppliers and buyers. For example, the use of platforms to facilitate communication and share data makes it possible to increase transparency, simplify communication, and speed up cooperation processes [82].
- **Integration of operations**
  The integration of the operations criterion in the supplier selection problem refers to the supplier's ability to use technologies such as IoT, GNSS, artificial intelligence, etc. to increase its efficiency and achieve integrated operations (Lo, 2024; [83]). The use of Industry 4.0-based technologies increases transparency, enables real-time monitoring of the process, and optimizes resource management ([84]; Sorooshain et al., 2024).
- **Security**
  Suppliers must provide a secure environment to protect their sensitive data. Industry 4.0-based technologies provide this possibility for suppliers. In general, the purpose of the security criterion is to measure the supplier's performance in protecting data, preventing cyber-attacks, preventing disruptions in processes, etc. (Lo, 2024; [83]).
- **Traceability**

The meaning of the traceability criterion in the supplier selection problem is to provide an environment for the real-time tracking of materials and continuous monitoring of processes [85]. This criterion increases transparency, prevents fraud, and increases buyers' satisfaction [86]. Technologies such as RFID-tags, closed-circuit television (CCTV), and vehicles equipped with GNSS are useful for tracking products and processes.

4. Proposed approach

The approach presented in this article is an extension of the GBWM provided by Tavana et al. [31]. By integrating GBWM, interval numbers, and Supermatrix structure, this paper provides an integrated practical approach for calculating factor weights and prioritizing alternatives in intertwined networks that have a hierarchical structure. The proposed approach is shown in Fig. 1. Furthermore, the proposed approach is described step by step below:

Step 1: Structure the network. In other words, identify the components of the network. Network components can include factors, sub-factors, and alternatives. It is assumed that there are N factors and M alternatives, and each factor contains a number of sub-factors. Note that indices i and j denote factors i and j. Also, k_i and h represent the sub-

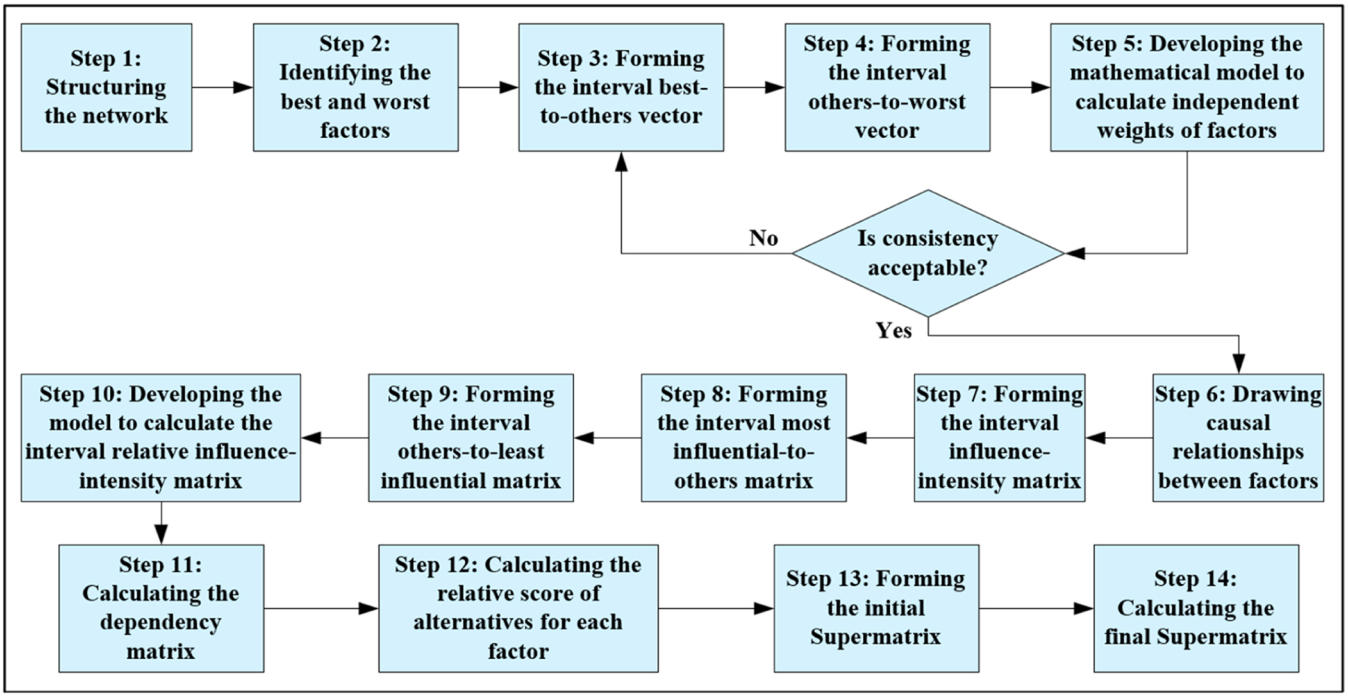


Fig. 1. The proposed approach.

factors of factor i and alternative h, respectively. Assuming the existence of a dependency among the factors, the overall structure of the investigated network is shown in Fig. 2.

Step 2: Determine the best and worst factors (sub-factors).

Step 3: Form the interval best-to-others vector by comparing the most important factor (sub-factor) with other factors (sub-factor) using the linguistic scales defined in Table 2. Let $[A_j^P, A_j^O]$ represent the interval best-to-others vector.

Step 4: Form the interval others-to-worst vector by comparing the least important factor (sub-factor) with other factors (sub-factor) using the linguistic scales defined in Table 2. Let $[B_j^P, B_j^O]$ represent the interval others-to-worst vector.

Step 5: Calculate the interval independent weights of factors (sub-factors) by expanding Model (1) and solving it.

$$Min\ \wp^* \tag{1}$$

$s.t.$

Table 2
Linguistic scales for pairwise comparisons.

Linguistic terms	Interval numbers
Equally significant	[1, 1]
Slightly significant	[2/3, 3/2]
Significant	[3/2, 5/2]
Very significant	[5/2, 7/2]
Absolutely significant	[7/2, 9/2]

$$\left|[\vartheta_B^P, \vartheta_B^O] - [A_j^P, A_j^O] \times [\vartheta_j^P, \vartheta_j^O]\right| \le [\wp^P, \wp^O] \forall j$$

$$\left|[\vartheta_j^P, \vartheta_j^O] - [B_j^P, B_j^O] \times [\vartheta_W^P, \vartheta_W^O]\right| \le [\wp^P, \wp^O] \forall j$$

$$\sum_j \frac{\vartheta_j^P + \vartheta_j^O}{2} = 1$$

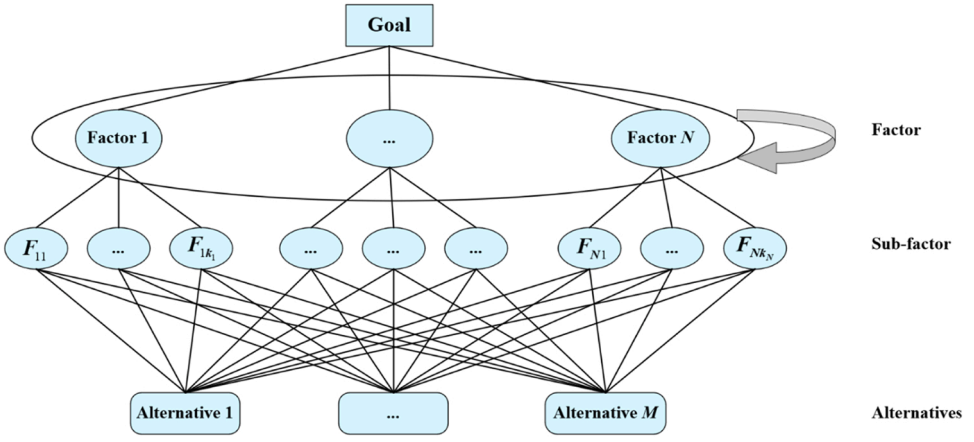


Fig. 2. The overall structure of the defined network.

$$\vartheta_j^P \le \vartheta_j^O$$
$$\vartheta_j^P \ge 0$$

where $[\wp^P, \wp^O]$ represents $Max\left\{\left|[\vartheta_B^P, \vartheta_B^O] - [A_j^P, A_j^O] \times [\vartheta_j^P, \vartheta_j^O]\right|, \left|[\vartheta_j^P, \vartheta_j^O] - [B_j^P, B_j^O] \times [\vartheta_W^P, \vartheta_W^O]\right|\right\}$. Also, $[\vartheta_B^P, \vartheta_B^O]$, $[\vartheta_W^P, \vartheta_W^O]$, and $[\vartheta_j^P, \vartheta_j^O]$ denote the interval weights of the best, worst, and jth factors, respectively. Considering $\wp^P \le \wp^O$, we suppose that the optimal value of objective function is $[\wp^*, \wp^*]$, such that $\wp^* \le \wp^P$. Therefore, Model (1) is converted to Model (2).

$$Min\ \wp^* \tag{2}$$

s.t.

$$\left|[\vartheta_B^P, \vartheta_B^O] - [A_j^P, A_j^O] \times [\vartheta_j^P, \vartheta_j^O]\right| \le [\wp^*, \wp^*] \forall j$$

$$\left|[\vartheta_j^P, \vartheta_j^O] - [B_j^P, B_j^O] \times [\vartheta_W^P, \vartheta_W^O]\right| \le [\wp^*, \wp^*] \forall j$$

$$\sum_j \frac{\vartheta_j^P + \vartheta_j^O}{2} = 1$$

$$\vartheta_j^P \le \vartheta_j^O$$
$$\vartheta_j^P \ge 0$$

Solving this model results in the optimal value of interval independent weights of factors (sub-factors) and objective function. If $\wp^* \le 0.1$, then there is consistency in pairwise comparisons [32]. But if $\wp^* > 0.1$, it means that the pairwise comparisons are inconsistent, and the vectors provided in steps 3 and 4 should be modified, and the model should be run again with modified data. Finally, the crisp independent weights of factors (sub-factors) are calculated using Eq. (3).

$$\vartheta_j^{CR} = \frac{\vartheta_j^P + \vartheta_j^O}{2} \forall j \tag{3}$$

Step 6: Present the cause-and-effect relationship between the factors in the form of a diagram.

Step 7: Form the interval influence-intensity matrix. For this purpose, it is required to determine the influence intensity of factors on each other by means of the linguistic scales defined in Table 3. If there is no causal relationship between two factors, use symbol "N". The "-" symbol is also placed on the main diameter of the matrix.

Step 8: Form the interval most influential-to-others matrix. To obtain this matrix, we should divide the largest number of each column of the interval influence-intensity matrix by all the numbers in that column. Let $[A_{F_j^i}^P, A_{F_j^i}^O]$ represent the column j of interval most influential-to-others matrix.

Step 9: Form the interval others-to-least influential matrix. To obtain this matrix, all the numbers of each column of the influence-intensity matrix should be divided by the smallest number of that column. Let $[B_{F_j^i}^P, B_{F_j^i}^O]$ represent the column j of interval others-to-least influential matrix.

Step 10: Calculate the interval relative influence-intensity weights by expanding Model (4) and solving it.

Table 3
The interval influence-intensity scales.

Linguistic terms	Interval numbers
No influence	[0, 0]
Slightly influence	[2/3, 3/2]
Influence	[3/2, 5/2]
Very influence	[5/2, 7/2]
Absolutely influence	[7/2, 9/2]

$$Min\ [\wp_j^P, \wp_j^O] \tag{4}$$

s.t.

$$\left|[\vartheta_{F_j^B}^P, \vartheta_{F_j^B}^O] - [A_{F_j^i}^P, A_{F_j^i}^O] \times [\vartheta_{F_j^i}^P, \vartheta_{F_j^i}^O]\right| \le [\wp_j^P, \wp_j^O] \forall j$$

$$\left|[\vartheta_{F_j^i}^P, \vartheta_{F_j^i}^O] - [B_{F_j^i}^P, B_{F_j^i}^O] \times [\vartheta_{F_j^W}^P, \vartheta_{F_j^W}^O]\right| \le [\wp_j^P, \wp_j^O] \forall j$$

$$\sum_i \frac{\vartheta_{F_j^i}^P + \vartheta_{F_j^i}^O}{2} = 1 \forall j$$

$$\vartheta_{F_j^i}^P \le \vartheta_{F_j^i}^O$$
$$\vartheta_{F_j^i}^P \ge 0$$

where $[\vartheta_{F_j^B}^P, \vartheta_{F_j^B}^O]$, $[\vartheta_{F_j^W}^P, \vartheta_{F_j^W}^O]$, and $[\vartheta_{F_j^i}^P, \vartheta_{F_j^i}^O]$ denote the interval relative influence-intensity weights of the best, worst, and ith factors in column j of the interval relative influence-intensity matrix. Considering $\wp_j^* \le \wp_j^P$, Model (4) can be transformed as follows:

$$Min\ \wp_j^* \tag{5}$$

s.t.

$$\left|[\vartheta_{F_j^B}^P, \vartheta_{F_j^B}^O] - [A_{F_j^i}^P, A_{F_j^i}^O] \times [\vartheta_{F_j^i}^P, \vartheta_{F_j^i}^O]\right| \le [\wp_j^*, \wp_j^*] \forall j$$

$$\left|[\vartheta_{F_j^i}^P, \vartheta_{F_j^i}^O] - [B_{F_j^i}^P, B_{F_j^i}^O] \times [\vartheta_{F_j^W}^P, \vartheta_{F_j^W}^O]\right| \le [\wp_j^*, \wp_j^*] \forall j$$

$$\sum_i \frac{\vartheta_{F_j^i}^P + \vartheta_{F_j^i}^O}{2} = 1 \forall j$$

$$\vartheta_{F_j^i}^P \le \vartheta_{F_j^i}^O$$
$$\vartheta_{F_j^i}^P \ge 0$$

By solving Model (5), the interval relative influence-intensity weights of the factors are calculated. Then, the interval relative influence-intensity matrix is formed by employing these weights. It is notable that in forming this matrix, we should put interval numbers [1,1] and [0,0] instead of the symbols "-" and "N", respectively.

Step 11: Normalize the matrix achieved from previous step via Eq. (6). Suppose that $\left[\alpha_{F_j^i}^P, \alpha_{F_j^i}^O\right]$ represents the interval number located in row i and column j of the normalized matrix. The matrix obtained from this step is called interval dependency matrix. Finally, the interval numbers of this matrix are converted into crisp numbers using Eq. (7).

$$\left[\alpha_{F_j^i}^P, \alpha_{F_j^i}^O\right] = \left[\frac{\vartheta_{F_j^i}^P}{\sum_i \frac{\vartheta_{F_j^i}^P + \vartheta_{F_j^i}^O}{2}}, \frac{\vartheta_{F_j^i}^O}{\sum_i \frac{\vartheta_{F_j^i}^P + \vartheta_{F_j^i}^O}{2}}\right] \tag{6}$$

$$\alpha_{F_j^i}^{CR} = \frac{\alpha_{F_j^i}^P + \alpha_{F_j^i}^O}{2} \forall i,j \tag{7}$$

Step 12: Calculate the relative score of the alternatives for each sub-factor. For this purpose, perform steps 2–5 on the alternatives for each sub-factor. In other words, for each sub-factor, select the best and worst alternatives (step 2); form the best-to-others and others-to-worst vectors for them (steps 3 and 4); and finally, calculate the score of each

alternative for each sub-factor (step 5). Suppose that $\left[\beta^{P}_{hk_j}, \beta^{O}_{hk_j}\right]$ represents the interval score of alternative h in sub-factor k_j. By applying Eq. (8), the crisp score of alternative h in sub-factor k_j is calculated.

$$\beta^{CR}_{hk_j} = \frac{\beta^{P}_{hk_j} + \beta^{O}_{hk_j}}{2} \forall h, k_j \tag{8}$$

Step 13: Form the initial Supermatrix (C) through the structure defined in Fig. 3. In the initial Supermatrix, $\left[\vartheta^{CR}_{i}\right]$ shows the matrix corresponding to the weights of the factors. In this Supermatrix, the interdependencies between the factors is represented by $\left[\alpha^{CR}_{F^i_j}\right]$. In addition, the weights of the sub-factors of each factor and the interdependencies between the sub-factors are denoted by $\left[\vartheta^{CR}_{k_i}\right]$ and $\left[\alpha^{CR}_{F^{k_i}_{k_j}}\right]$, respectively. Finally, $\left[\beta^{CR}_{hk_i}\right]$ demonstrates the score of each alternative for each subfactor.

Step 14: Determine the final Supermatrix (D) via Eq. (9). Note It should be considered that the final Supermatrix provides the final (dependent) weights of factors and sub-factors, and the final relative scores of alternatives.

$$D = \lim_{e \to \infty} S^{e} \tag{9}$$

It is worth noting that the brainstorming method is used to identify the best and worst factors and subfactors, fill out paired comparison questionnaires, and determine the influence intensity of factors on each other. This method is applied to converge and reach a consensus among experts.

5. Case study

In order to check the applicability of the proposed approach, we rank the suppliers of a construction company in Malaysia through the developed approach. Company Y is a Malaysian mass construction company that works in the field of building towers, residential towns, shopping malls, etc. So far, this company has implemented more than 55 successful projects in Malaysia. Recently, this company has built a residential tower worth approximately 124 million dollars in Southeast Malaysia. In this article, the knowledge of four experts of this company is applied to rank six potential suppliers of this project from the circularity and Industry 5.0 perspectives through the developed approach. In doing so, the experts with at least eight years of professional experience in the field of construction were selected while they were in charge of the management position in the studied project. The demographic information of the experts is presented in Table 4. It should be noted that data collection was conducted through expert-administered questionnaires, ensuring consensus among all experts. The implementation steps of the developed approach are given below:

Step 1: This step deals with identifying the factors and sub-factors of circular supplier 5.0 evaluation. In order to consider industry 5.0 perspective, the criteria of sustainability, human-centricity, and resiliency should be included in the evaluation process [87]. On the other hand, sustainability measures include economic, environmental, and social dimensions. Therefore, with circularity vision, supplier 5.0 evaluation factors include resiliency, human-centricity, economic, social, circular, and Industry 4.0. After identifying six evaluation factors, a complete set of criteria extracted from the literature was provided to the experts, and based on their consensus, sub-factors of each factor were identified. The sub-factors related to each factor are listed in Table 5.

	Goal	Factor	Sub-factor	Alternative
Goal	0	0	0	0
Factor	$\left[\vartheta^{CR}_{i}\right]$	$\left[\alpha^{CR}_{F^i_j}\right]$	[0]	[0]
Sub-factor	[0]	$\left[\vartheta^{CR}_{k_i}\right]$	$\left[\alpha^{CR}_{F^{k_i}_{k_j}}\right]$	[0]
Alternative	[0]	[0]	$\left[\beta^{CR}_{hk_i}\right]$	[0]

Fig. 3. Overall structure of the initial Supermatrix.

Table 4
Demographic information of experts.

	No. of experts
Gender	
Male	4
Female	0
Age	
40–50 years	3
Above 50 years	1
Experience	
8–10 years	1
10–15 years	2
Above 15 years	1
Expertise	
Project manager	1
Project control manager	1
Procurement manager	1
Construction manager	1

Step 2: In this step, the best and worst factors and sub-factors were determined by experts, as displayed in Table 6.

Step 3: In this step, the best-to-others vectors are formed for factors and sub-factors. Table 7 shows this vector for factors. Tables A1 to A6 represent this vector for the sub-factors.

Step 4: In this step, the others-to-worst vectors are formed for factors and sub-factors. Table 8 shows this vector for factors. Tables A7 to A12 represent this vector for the sub-factors.

Step 5: In this step, Model (2) is developed by using the data provided in steps 3 and 4. For example, the model developed for the factors using the data in Tables 7 and 8 is given below.

$$Min\ \wp^{*}$$

$$s.t.$$

$$\left|\left[\vartheta^{P}_{E}, \vartheta^{O}_{E}\right] - \left[\frac{2}{3}, \frac{3}{2}\right] \times \left[\vartheta^{P}_{R}, \vartheta^{O}_{R}\right]\right| \le \left[\wp^{*}, \wp^{*}\right]$$

$$\left|\left[\vartheta^{P}_{E}, \vartheta^{O}_{E}\right] - \left[\frac{3}{2}, \frac{5}{2}\right] \times \left[\vartheta^{P}_{H}, \vartheta^{O}_{H}\right]\right| \le \left[\wp^{*}, \wp^{*}\right]$$

$$\left|\left[\vartheta^{P}_{E}, \vartheta^{O}_{E}\right] - \left[\frac{7}{2}, \frac{9}{2}\right] \times \left[\vartheta^{P}_{S}, \vartheta^{O}_{S}\right]\right| \le \left[\wp^{*}, \wp^{*}\right]$$

$$\left|\left[\vartheta^{P}_{E}, \vartheta^{O}_{E}\right] - \left[\frac{5}{2}, \frac{7}{2}\right] \times \left[\vartheta^{P}_{C}, \vartheta^{O}_{C}\right]\right| \le \left[\wp^{*}, \wp^{*}\right]$$

$$\left|\left[\vartheta^{P}_{E}, \vartheta^{O}_{E}\right] - \left[\frac{5}{2}, \frac{7}{2}\right] \times \left[\vartheta^{P}_{I}, \vartheta^{O}_{I}\right]\right| \le \left[\wp^{*}, \wp^{*}\right]$$

$$\left|\left[\vartheta^{P}_{R}, \vartheta^{O}_{R}\right] - \left[\frac{5}{2}, \frac{7}{2}\right] \times \left[\vartheta^{P}_{S}, \vartheta^{O}_{S}\right]\right| \le \left[\wp^{*}, \wp^{*}\right]$$

$$\left|\left[\vartheta^{P}_{H}, \vartheta^{O}_{H}\right] - \left[\frac{3}{2}, \frac{5}{2}\right] \times \left[\vartheta^{P}_{S}, \vartheta^{O}_{S}\right]\right| \le \left[\wp^{*}, \wp^{*}\right]$$

$$\left|\left[\vartheta^{P}_{C}, \vartheta^{O}_{C}\right] - \left[\frac{2}{3}, \frac{3}{2}\right] \times \left[\vartheta^{P}_{S}, \vartheta^{O}_{S}\right]\right| \le \left[\wp^{*}, \wp^{*}\right]$$

Table 5
Factors and sub-factors of evaluating the circular supplier 5.0.

Factor	Sub-factor	Description
Resiliency (R)	Backup suppliers (R1)	Applying the strategy of cooperating with backup suppliers
	Flexibility and agility (R2)	Adopting methods to increase flexibility and agility in response to changing demands
	Multiple sources (R3)	Adopting a multiple sourcing strategy instead of single one
	Data-driven analytical methods (R4)	Using data-driven analytical methods for identifying risks and mitigating them
Human-centricity (H)	Workforce health and safety (H1)	Adopting safety and health standard for protecting the workforce
	Ethical labor practices (H2)	Avoiding unethical labor practices such as forced and child labor
	Human rights (H3)	Protecting the rights of workforce and shareholders
	Employee training (H4)	Adopting strategies to train the workforce and assign appropriate tasks to them
Economic (E)	Cost (E1)	Using mechanisms to reduce the cost of products
	Delivery (E2)	Employing mechanisms to increase speed, accuracy, and reliability of delivery
	Quality (E3)	Applying strategies to increase the quality of products and reduce their rejection rate
	Capability (E4)	Technological and financial capabilities
	Reputation (E5)	Financial and performance history
Social (S)	Information sharing (S1)	Adopting strategies to share supplier information and data with buyers
	Diversity and inclusion (S2)	Paying attention to diversity and inclusion programs
	Employment practices (S3)	Use of appropriate mechanisms in employment practices
	Job creation (S4)	Paying attention to job creation
Circular (C)	Energy consumption (C1)	Applying practices to improve energy consumption in production and recycling process
	Carbon emission (C2)	Employing methods to mitigate the carbon footprint in production and recycling
	Clean technologies (C3)	Adopting eco-friendly and clean technologies
	Waste management (C4)	Adopting methods to manage waste
	Green R&D (C5)	Green innovation and R&D
	Eco-friendly packaging (C6)	Adopting eco-friendly and recyclable materials in packaging
Industry 4.0 (I)	Platforms (I1)	Adopting platforms to facilitate communication and share data
	Integration of operations (I2)	Applying Industry 4.0-based technologies such as IoT to integrate operations
	Security (I3)	Adopting Industry 4.0-based technologies for protecting sensitive data and information
	Traceability (I4)	Adopting Industry 4.0-based technologies to track materials and monitor logistics and inventory

Table 6
The best and worst factors and sub-factors.

	Best	Worst
Factor	Economic	Social
Resiliency	R2	R1
Human-centricity	H1	H2
Economic	E1	E2
Social	S1	S2
Circular	C2	C5
Industry 4.0	I3	I2

$$\left|\left[\vartheta_I^P, \vartheta_I^O\right] - \left[\frac{2}{3}, \frac{3}{2}\right] \times \left[\vartheta_S^P, \vartheta_S^O\right]\right| \leq \left[\wp^*, \wp^*\right]$$

$$\frac{\vartheta_R^P + \vartheta_R^O}{2} + \frac{\vartheta_H^P + \vartheta_H^O}{2} + \frac{\vartheta_E^P + \vartheta_E^O}{2} + \frac{\vartheta_S^P + \vartheta_S^O}{2} + \frac{\vartheta_C^P + \vartheta_C^O}{2} + \frac{\vartheta_I^P + \vartheta_I^O}{2} = 1$$

$$\vartheta_R^P \leq \vartheta_R^O; \vartheta_H^P \leq \vartheta_H^O; \vartheta_E^P \leq \vartheta_E^O; \vartheta_S^P \leq \vartheta_S^O; \vartheta_C^P \leq \vartheta_C^O; \vartheta_I^P \leq \vartheta_I^O$$
$$\vartheta_R^P, \vartheta_H^P, \vartheta_E^P, \vartheta_S^P, \vartheta_C^P, \vartheta_I^P \geq 0$$

We can expand the model as follows:

$Min\ \wp^*$

s.t.

$$\vartheta_E^P \leq \frac{2}{3} \times \vartheta_R^P + \wp^*; \vartheta_E^O \leq \frac{3}{2} \times \vartheta_R^O + \wp^*; \vartheta_E^P \geq \frac{2}{3} \times \vartheta_R^P - \wp^*; \vartheta_E^O \geq \frac{3}{2} \times \vartheta_R^O - \wp^*$$

$$\vartheta_E^P \leq \frac{3}{2} \times \vartheta_H^P + \wp^*; \vartheta_E^O \leq \frac{5}{2} \times \vartheta_H^O + \wp^*; \vartheta_E^P \geq \frac{3}{2} \times \vartheta_H^P - \wp^*; \vartheta_E^O \geq \frac{5}{2} \times \vartheta_H^O - \wp^*$$

$$\vartheta_E^P \leq \frac{7}{2} \times \vartheta_S^P + \wp^*; \vartheta_E^O \leq \frac{9}{2} \times \vartheta_S^O + \wp^*; \vartheta_E^P \geq \frac{7}{2} \times \vartheta_S^P - \wp^*; \vartheta_E^O \geq \frac{9}{2} \times \vartheta_S^O - \wp^*$$

$$\vartheta_E^P \leq \frac{5}{2} \times \vartheta_C^P + \wp^*; \vartheta_E^O \leq \frac{7}{2} \times \vartheta_C^O + \wp^*; \vartheta_E^P \geq \frac{5}{2} \times \vartheta_C^P - \wp^*; \vartheta_E^O \geq \frac{7}{2} \times \vartheta_C^O - \wp^*$$

$$\vartheta_E^P \leq \frac{5}{2} \times \vartheta_I^P + \wp^*; \vartheta_E^O \leq \frac{7}{2} \times \vartheta_I^O + \wp^*; \vartheta_E^P \geq \frac{5}{2} \times \vartheta_I^P - \wp^*; \vartheta_E^O \geq \frac{7}{2} \times \vartheta_I^O - \wp^*$$

$$\vartheta_R^P \leq \frac{5}{2} \times \vartheta_S^P + \wp^*; \vartheta_R^O \leq \frac{7}{2} \times \vartheta_S^O + \wp^*; \vartheta_R^P \geq \frac{5}{2} \times \vartheta_S^P - \wp^*; \vartheta_R^O \geq \frac{7}{2} \times \vartheta_S^O - \wp^*$$

$$\vartheta_H^P \leq \frac{3}{2} \times \vartheta_S^P + \wp^*; \vartheta_H^O \leq \frac{5}{2} \times \vartheta_S^O + \wp^*; \vartheta_H^P \geq \frac{3}{2} \times \vartheta_S^P - \wp^*; \vartheta_H^O \geq \frac{5}{2} \times \vartheta_S^O - \wp^*$$

$$\vartheta_C^P \leq \frac{2}{3} \times \vartheta_S^P + \wp^*; \vartheta_C^O \leq \frac{3}{2} \times \vartheta_S^O + \wp^*; \vartheta_C^P \geq \frac{2}{3} \times \vartheta_S^P - \wp^*; \vartheta_C^O \geq \frac{3}{2} \times \vartheta_S^O - \wp^*$$

$$\vartheta_I^P \leq \frac{2}{3} \times \vartheta_S^P + \wp^*; \vartheta_I^O \leq \frac{3}{2} \times \vartheta_S^O + \wp^*; \vartheta_I^P \geq \frac{2}{3} \times \vartheta_S^P - \wp^*; \vartheta_I^O \geq \frac{3}{2} \times \vartheta_S^O - \wp^*$$

$$\frac{\vartheta_R^P + \vartheta_R^O}{2} + \frac{\vartheta_H^P + \vartheta_H^O}{2} + \frac{\vartheta_E^P + \vartheta_E^O}{2} + \frac{\vartheta_S^P + \vartheta_S^O}{2} + \frac{\vartheta_C^P + \vartheta_C^O}{2} + \frac{\vartheta_I^P + \vartheta_I^O}{2} = 1$$

$$\vartheta_R^P \leq \vartheta_R^O; \vartheta_H^P \leq \vartheta_H^O; \vartheta_E^P \leq \vartheta_E^O; \vartheta_S^P \leq \vartheta_S^O; \vartheta_C^P \leq \vartheta_C^O; \vartheta_I^P \leq \vartheta_I^O$$
$$\vartheta_R^P, \vartheta_H^P, \vartheta_E^P, \vartheta_S^P, \vartheta_C^P, \vartheta_I^P \geq 0$$

By running the obtained model in CPLEX solver through GAMS software, the interval independent weights of the factors and the inconsistency rate (i.e., $\wp^*$) are calculated, as given in Tables 9,10 Similarly, we implement these operations on the sub-factors of each factor. Table 9 shows the results of these operations. It is noteworthy that the crisp independent weights of factors and sub-factors are calculated by Eq. (3).

Step 6: In this step, experts drew cause-and-effect diagram for the factors. Fig. 4 represents this diagram.

Step 7: In this step, experts determine the influence intensity of the factors on each other based on Fig. 4 and using the linguistic scales defined in Table 3.

Step 8: Based on the instructions described in step 8 of the proposed approach section, the interval most influential-to-others matrix is formed. Table 11 represents this matrix.

Step 9: Based on the instructions described in step 9 of the proposed approach section, the interval others-to-least influential matrix is formed. Table 12 represents this matrix.

Table 7
The best to-others vector for factors.

	Resiliency	Human-centricity	Economic	Social	Circular	Industry 4.0
Best: Economic	[2/3, 3/2]	[3/2, 5/2]	[1, 1]	[7/2, 9/2]	[5/2, 7/2]	[5/2, 7/2]

Table 8
The others-to-worst vector for factors.

	Worst: Social
Resiliency	[5/2, 7/2]
Human-centricity	[3/2, 5/2]
Economic	[7/2, 9/2]
Social	[1, 1]
Circular	[2/3, 3/2]
Industry 4.0	[2/3, 3/2]

Step 10: In this step, using the data of Tables 11 and 12, Model (5) is developed. For example, based on the first column of these two tables, the model is developed as follows.

$$Min\ \wp_R^*$$

$$s.t.$$

$$\left|\left[\vartheta^P_{F^E_R},\vartheta^O_{F^E_R}\right]-\left[1,\frac{7}{3}\right]\times\left[\vartheta^P_{F^H_R},\vartheta^O_{F^H_R}\right]\right|\le\left[\wp^*_R,\wp^*_R\right]$$

$$\left|\left[\vartheta^P_{F^E_R},\vartheta^O_{F^E_R}\right]-\left[\frac{5}{3},\frac{21}{4}\right]\times\left[\vartheta^P_{F^S_R},\vartheta^O_{F^S_R}\right]\right|\le\left[\wp^*_R,\wp^*_R\right]$$

$$\left|\left[\vartheta^P_{F^E_R},\vartheta^O_{F^E_R}\right]-\left[1,\frac{7}{3}\right]\times\left[\vartheta^P_{F^I_R},\vartheta^O_{F^I_R}\right]\right|\le\left[\wp^*_R,\wp^*_R\right]$$

$$\left|\left[\vartheta^P_{F^H_R},\vartheta^O_{F^H_R}\right]-\left[1,\frac{15}{4}\right]\times\left[\vartheta^P_{F^S_R},\vartheta^O_{F^S_R}\right]\right|\le\left[\wp^*_R,\wp^*_R\right]$$

$$\left|\left[\vartheta^P_{F^I_R},\vartheta^O_{F^I_R}\right]-\left[1,\frac{15}{4}\right]\times\left[\vartheta^P_{F^S_R},\vartheta^O_{F^S_R}\right]\right|\le\left[\wp^*_R,\wp^*_R\right]$$

$$\frac{\vartheta^P_{F^E_R}+\vartheta^O_{F^E_R}}{2}+\frac{\vartheta^P_{F^H_R}+\vartheta^O_{F^H_R}}{2}+\frac{\vartheta^P_{F^S_R}+\vartheta^O_{F^S_R}}{2}+\frac{\vartheta^P_{F^I_R}+\vartheta^O_{F^I_R}}{2}=1$$

$$\vartheta^P_{F^E_R}\le\vartheta^O_{F^E_R};\vartheta^P_{F^H_R}\le\vartheta^O_{F^H_R};\vartheta^P_{F^S_R}\le\vartheta^O_{F^S_R};\vartheta^P_{F^I_R}\le\vartheta^O_{F^I_R}$$
$$\vartheta^P_{F^E_R},\vartheta^P_{F^H_R},\vartheta^P_{F^S_R},\vartheta^P_{F^I_R}\ge 0$$

We can expand the model as follows:

$$Min\ \wp_R^*$$

Table 9
The interval and crisp independent weights of factors and sub-factors.

Factor ($\wp^* = 0.037$)	Interval independent weight	Crisp independent weight	Sub-factor	Interval independent weight	Crisp independent weight
Resiliency ($\wp^* = 0.015$)	[0.170, 0.331]	0.2500	R1	[0.027, 0.216]	0.1215
			R2	[0.053, 0.772]	0.4125
			R3	[0.034, 0.316]	0.1750
			R4	[0.057, 0.525]	0.2910
Human-centricity ($\wp^* = 0.015$)	[0.116, 0.199]	0.1575	H1	[0.053, 0.772]	0.4125
			H2	[0.027, 0.216]	0.1215
			H3	[0.057, 0.525]	0.2910
			H4	[0.034, 0.316]	0.1750
Economic ($\wp^* = 0.041$)	[0.150, 0.460]	0.3050	E1	[0.168, 0.514]	0.3410
			E2	[0.060, 0.105]	0.0825
			E3	[0.190, 0.371]	0.2805
			E4	[0.130, 0.222]	0.1760
			E5	[0.081, 0.159]	0.1200
Social ($\wp^* = 0.047$)	[0.053, 0.094]	0.0735	S1	[0.190, 0.584]	0.3870
			S2	[0.068, 0.120]	0.0940
			S3	[0.148, 0.253]	0.2005
			S4	[0.216, 0.421]	0.3185
Circular ($\wp^* = 0.035$)	[0.072, 0.142]	0.1070	C1	[0.111, 0.189]	0.1500
			C2	[0.142, 0.438]	0.2900
			C3	[0.161, 0.315]	0.2380
			C4	[0.069, 0.135]	0.1020
			C5	[0.050, 0.090]	0.0700
			C6	[0.111, 0.189]	0.1500
Industry 4.0 ($\wp^* = 0.014$)	[0.072, 0.142]	0.1070	I1	[0.035, 0.355]	0.1950
			I2	[0.032, 0.246]	0.1390
			I3	[0.066, 0.876]	0.4710
			I4	[0.035, 0.355]	0.1950

Table 10
The interval influence intensity matrix.

	Resiliency	Human-centricity	Economic	Social	Circular	Industry 4.0
Resiliency	-	[3/2, 5/2]	N	N	N	N
Human-centricity	[3/2, 5/2]	-	[2/3, 3/2]	[2/3, 3/2]	N	N
Economic	[5/2, 7/2]	[3/2, 5/2]	-	N	[3/2, 5/2]	[2/3, 3/2]
Social	[2/3, 3/2]	[3/2, 5/2]	[3/2, 5/2]	-	[2/3, 3/2]	N
Circular	N	[2/3, 3/2]	[2/3, 3/2]	N	-	N
Industry 4.0	[3/2, 5/2]	[2/3, 3/2]	[3/2, 5/2]	[3/2, 5/2]	[2/3, 3/2]	-

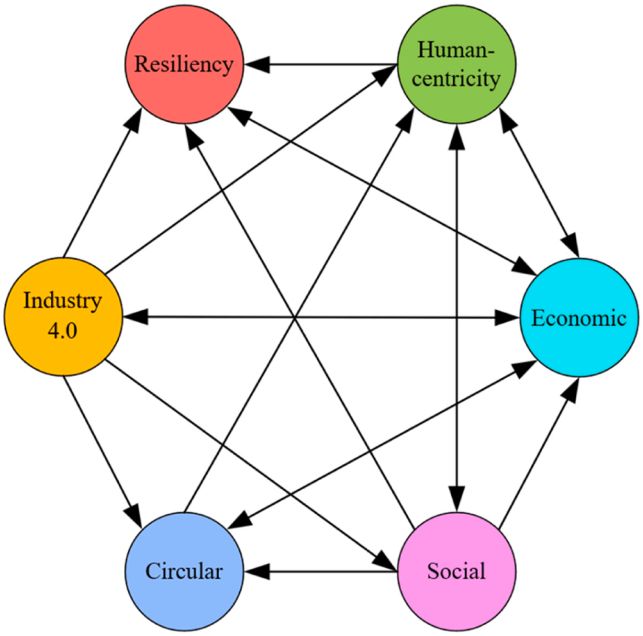


Fig. 4. The cause-and-effect diagram for factors.

s.t.

$$\vartheta^{P}_{F^{E}_{R}} \le \vartheta^{P}_{F^{H}_{R}} + \wp^{*}_{R}; \vartheta^{O}_{F^{E}_{R}} \le \frac{7}{3} \times \vartheta^{O}_{F^{H}_{R}} + \wp^{*}_{R}; \vartheta^{P}_{F^{E}_{R}} \ge \vartheta^{P}_{F^{H}_{R}} - \wp^{*}_{R}; \vartheta^{O}_{F^{E}_{R}} \ge \frac{7}{3} \times \vartheta^{O}_{F^{H}_{R}} - \wp^{*}_{R}$$

$$\vartheta^{P}_{F^{E}_{R}} \le \frac{5}{3} \times \vartheta^{P}_{F^{S}_{R}} + \wp^{*}_{R}; \vartheta^{O}_{F^{E}_{R}} \le \frac{21}{4} \times \vartheta^{O}_{F^{S}_{R}} + \wp^{*}_{R}; \vartheta^{P}_{F^{E}_{R}} \ge \frac{5}{3} \times \vartheta^{P}_{F^{S}_{R}} - \wp^{*}_{R}; \vartheta^{O}_{F^{E}_{R}} \ge \frac{21}{4} \times \vartheta^{O}_{F^{S}_{R}} - \wp^{*}_{R}$$

$$\vartheta^{P}_{F^{E}_{R}} \le \vartheta^{P}_{F^{I}_{R}} + \wp^{*}_{R}; \vartheta^{O}_{F^{E}_{R}} \le \frac{7}{3} \times \vartheta^{O}_{F^{I}_{R}} + \wp^{*}_{R}; \vartheta^{P}_{F^{E}_{R}} \ge \vartheta^{P}_{F^{I}_{R}} - \wp^{*}_{R}; \vartheta^{O}_{F^{E}_{R}} \ge \frac{7}{3} \times \vartheta^{O}_{F^{I}_{R}} - \wp^{*}_{R}$$

$$\vartheta^{P}_{F^{H}_{R}} \le \vartheta^{P}_{F^{S}_{R}} + \wp^{*}_{R}; \vartheta^{O}_{F^{H}_{R}} \le \frac{15}{4} \times \vartheta^{O}_{F^{S}_{R}} + \wp^{*}_{R}; \vartheta^{P}_{F^{H}_{R}} \ge \vartheta^{P}_{F^{S}_{R}} - \wp^{*}_{R}; \vartheta^{O}_{F^{H}_{R}} \ge \frac{15}{4} \times \vartheta^{O}_{F^{S}_{R}} - \wp^{*}_{R}$$

$$\vartheta^{P}_{F^{I}_{R}} \le \vartheta^{P}_{F^{S}_{R}} + \wp^{*}_{R}; \vartheta^{O}_{F^{I}_{R}} \le \frac{15}{4} \times \vartheta^{O}_{F^{S}_{R}} + \wp^{*}_{R}; \vartheta^{P}_{F^{I}_{R}} \ge \vartheta^{P}_{F^{S}_{R}} - \wp^{*}_{R}; \vartheta^{O}_{F^{I}_{R}} \ge \frac{15}{4} \times \vartheta^{O}_{F^{S}_{R}} - \wp^{*}_{R}$$

$$\frac{\vartheta^{P}_{F^{E}_{R}} + \vartheta^{O}_{F^{E}_{R}}}{2} + \frac{\vartheta^{P}_{F^{H}_{R}} + \vartheta^{O}_{F^{H}_{R}}}{2} + \frac{\vartheta^{P}_{F^{S}_{R}} + \vartheta^{O}_{F^{S}_{R}}}{2} + \frac{\vartheta^{P}_{F^{I}_{R}} + \vartheta^{O}_{F^{I}_{R}}}{2} = 1$$

$$\vartheta^{P}_{F^{E}_{R}} \le \vartheta^{O}_{F^{E}_{R}}; \vartheta^{P}_{F^{H}_{R}} \le \vartheta^{O}_{F^{H}_{R}}; \vartheta^{P}_{F^{S}_{R}} \le \vartheta^{O}_{F^{S}_{R}}; \vartheta^{P}_{F^{I}_{R}} \le \vartheta^{O}_{F^{I}_{R}}$$

$$\vartheta^{P}_{F^{E}_{R}}, \vartheta^{P}_{F^{H}_{R}}, \vartheta^{P}_{F^{S}_{R}}, \vartheta^{P}_{F^{I}_{R}} \ge 0$$

By running the obtained model in CPLEX solver through GAMS software, the interval relative influence-intensity weights and the inconsistency rate are calculated, which are given in Table 13. Similarly, we implement these operations on other columns of Tables 11 and 12. Table 13 shows the results of these operations.

Step 11: In this step, the matrix shown in Table 13 is first normalized by Eq. (6) and, then, the interval numbers are converted into crisp numbers by applying Eq. (7). The dependency matrix with interval and crisp numbers are represented in Tables 14 and 15.

Step 12: In this step, the relative score of suppliers is determined for each sub-factor. For this purpose, for each sub-factor, the best supplier is compared with other suppliers using the linguistic scales defined in Table 2 to obtain the best-to-others vector, as shown in Table B1. Similarly, for each sub-factor, the worst supplier is compared with other suppliers using the linguistic scales defined in Table 2 to obtain the others-to-worst vector, as shown in Table B2. Then, Model (2) is utilized to calculate the interval relative scores of suppliers for each sub-factor. Finally, the interval scores are converted to crisp scores by applying Eq. (3). Table 16 represents the interval and crisp relative scores of suppliers for each sub-factor.

Step 13: In this step, the initial Supermatrix is formed using the structure described in Fig. 3. Table B3 displays the initial Supermatrix.

Step 14: In this step, the final Supermatrix is obtained through Eq. (9). It should be noted that we can achieve the Supermatrix for $d \ge 23$, as shown in Table B4.

The Supermatrix shown in Table B4 provides the dependent (final) weights of factors and sub-factors, and the relative score of suppliers. Fig. 5 depicts the final weights of the sub-factors. The results presented in this table indicate that "economic" and "resiliency" factors are the most and least important factors, respectively. Also, "security" and "backup suppliers" are identified as the most and least important sub-factors, respectively. In addition, it is inferred from this table that supplier 2 outperforms the other suppliers and is thereby selected as the best

Table 11
The interval most influential-to-others matrix.

	Resiliency	Human-centricity	Economic	Social	Circular	Industry 4.0
Resiliency	-	[3/5, 5/3]	N	N	N	N
Human-centricity	[1, 7/3]	-	[1, 15/4]	[1, 15/4]	N	N
Economic	[5/7, 7/5]	[3/5, 5/3]	-	N	[3/5, 5/3]	[4/9, 9/4]
Social	[5/3, 21/4]	[3/5, 5/3]	[3/5, 5/3]	-	[1, 15/4]	N
Circular	N	[1, 15/4]	[1, 15/4]	N	-	N
Industry 4.0	[1, 7/3]	[1, 15/4]	[3/5, 5/3]	[3/5, 5/3]	[1, 15/4]	-

Table 12
The interval others-to-least influential matrix.

	Resiliency	Human-centricity	Economic	Social	Circular	Industry 4.0
Resiliency	-	[1, 15/4]	N	N	N	N
Human-centricity	[1, 15/4]	-	[4/9, 9/4]	[4/9, 9/4]	N	N
Economic	[5/3, 21/4]	[1, 15/4]	-	N	[1, 15/4]	[4/9, 9/4]
Social	[4/9, 9/4]	[1, 15/4]	[1, 15/4]	-	[4/9, 9/4]	N
Circular	N	[4/9, 9/4]	[4/9, 9/4]	N	-	N
Industry 4.0	[1, 15/4]	[4/9, 9/4]	[1, 15/4]	[1, 15/4]	[4/9, 9/4]	-

Table 13
The interval relative influence-intensity matrix.

	Resiliency($\wp_R^* = 0.081$)	Human-centricity($\wp_H^* = 0$)	Economic($\wp_E^* = 0$)	Social($\wp_S^* = 0$)	Circular($\wp_C^* = 0$)	Industry 4.0($\wp_I^* = 0$)
Resiliency	[1, 1]	[0.110, 0.411]	[0, 0]	[0, 0]	[0, 0]	[0, 0]
Human-centricity	[0.180, 0.294]	[1, 1]	[0.148, 0.148]	[0, 0.421]	[0, 0]	[0, 0]
Economic	[0.247, 0.605]	[0.110, 0.411]	[1, 1]	[0, 0]	[0, 1.304]	[1, 1]
Social	[0.100, 0.100]	[0.110, 0.411]	[0.148, 0.556]	[1, 1]	[0, 0.348]	[0, 0]
Circular	[0, 0]	[0.110, 0.110]	[0.148, 0.148]	[0, 0]	[1, 1]	[0, 0]
Industry 4.0	[0.180, 0.294]	[0.110, 0.110]	[0.148, 0.556]	[0, 1.579]	[0, 0.348]	[1, 1]

Table 14
The interval dependency matrix.

	Resiliency	Human-centricity	Economic	Social	Circular	Industry 4.0
Resiliency	[0.500, 0.500]	[0.055, 0.2055]	[0, 0]	[0, 0]	[0, 0]	[0, 0]
Human-centricity	[0.090, 0.147]	[0.500, 0.500]	[0.074, 0.074]	[0, 0.2105]	[0, 0]	[0, 0]
Economic	[0.1235, 0.3025]	[0.055, 0.2055]	[0.500, 0.500]	[0, 0]	[0, 0.652]	[0.500, 0.500]
Social	[0.050, 0.050]	[0.055, 0.2055]	[0.074, 0.278]	[0.500, 0.500]	[0, 0.174]	[0, 0]
Circular	[0, 0]	[0.055, 0.055]	[0.074, 0.074]	[0, 0]	[0.500, 0.500]	[0, 0]
Industry 4.0	[0.090, 0.147]	[0.055, 0.055]	[0.074, 0.278]	[0, 0.7895]	[0, 0.174]	[0.500, 0.500]

Table 15
The dependency matrix.

	Resiliency	Human-centricity	Economic	Social	Circular	Industry 4.0
Resiliency	0.5000	0.1300	0	0	0	0
Human-centricity	0.1185	0.5000	0.0740	0.1052	0	0
Economic	0.2130	0.1300	0.5000	0	0.3260	0.5000
Social	0.0500	0.1300	0.1760	0.5000	0.0870	0
Circular	0	0.0550	0.0740	0	0.5000	0
Industry 4.0	0.1185	0.0550	0.1760	0.3948	0.0870	0.5000

supplier.

Having obtained the results, we need to ensure that they are reliable. One of the main challenges in MCDM problems is collecting input data from experts. Experts may make errors for various reasons, such as rushing to fill out questionnaires, lack of concentration, etc. If the final results are highly sensitive to the data collected from experts, experts must be very careful to fill out questionnaires. However, in practice, it is very difficult to gain perfect information from experts. Developing a robust approach that is not highly sensitive to input data is a suitable solution to address this challenge. In this study, to examine the robustness of the proposed approach, we have defined six scenarios with changes in the input data provided by experts and implemented the proposed approach for each scenario. The results obtained for the scenarios are given in Tables 17 and 18. These scenarios are given below:

- **Scenario 1:** This scenario expands the range of interval numbers for the best-to-others and others-to-worst vectors related to the factors and sub-factors. To this end, the right-hand values (upper bounds) of the interval numbers presented in Tables 6, 7, and A1 to A12 are multiplied by 1.1, while the left-hand values (lower bounds) are multiplied by 0.9. By implementing the proposed approach on the new data, the final weights of the sub-factors and supplier scores are obtained, which are given in Tables 17 and 18, respectively.
- **Scenario 2:** This scenario decreases the range of interval numbers for the best-to-others and others-to-worst vectors related to the factors and sub-factors. In doing so, the right-hand values of the interval numbers provided in Tables 6, 7, and A1 to A12 are multiplied by 0.9, while the left-hand values are multiplied by 1.1. By implementing the proposed approach on the new data, the final weights of the sub-factors and supplier scores are obtained, as shown in Tables 17 and 18, respectively.
- **Scenario 3:** This scenario expands the range of interval numbers for the best-to-others and others-to-worst vectors related to the alternatives. In this vein, the right-hand values of the interval numbers provided in Tables B1 and B2 are multiplied by 1.1, while the left-hand values are multiplied by 0.9. By implementing the proposed approach on the new data, the final weights of the sub-factors and supplier scores are obtained, which are given in Tables 17 and 18, respectively.
- **Scenario 4:** This scenario decreases the range of interval numbers for the best-to-others and others-to-worst vectors related to the alternatives. In this vein, the right-hand values of the interval numbers presented in Tables B1 and B2 are multiplied by 0.9, while the left-hand values are multiplied by 1.1. By implementing the proposed approach on the new data, the final weights of the sub-factors and supplier scores are obtained, which are given in Tables 17 and 18, respectively.
- **Scenario 5:** This scenario consists of combining scenarios 1 and 3. In other words, in this scenario, the range of interval numbers for the best-to-others and others-to-worst vectors related to the factors, sub-factors, and alternatives is expanded. For this purpose, the right-hand values of the interval numbers provided in Tables 6, 7, A1 to A12, B1, and B2 are multiplied by 1.1, while the left-hand values are multiplied by 0.9. By implementing the proposed approach on the new data, the final weights of the sub-factors and supplier scores are obtained, which are given in Tables 17 and 18, respectively.
- **Scenario 6:** This scenario consists of combining scenarios 2 and 4. In other words, in this scenario, the range of interval numbers for the best-to-others and others-to-worst vectors related to the factors, sub-factors, and alternatives is reduced. For this purpose, the right-hand values of the interval numbers provided in Tables 6, 7, A1 to A12, B1, and B2 are multiplied by 0.9, while the left-hand values are multiplied by 1.1. By implementing the proposed approach on the new data, the final weights of the sub-factors and supplier scores are obtained, which are given in Tables 17 and 18, respectively. Also, Fig. 6 denotes the rank of suppliers in each scenario.

As shown in Table 17, changing the range of interval numbers does

Table 16
Relative scores of suppliers for each sub-factor.

Sub-factor	SU1		SU2		SU3		SU4		SU5		SU6	
	Interval score	Crisp score	Interval score	Crisp score	Interval score	Crisp score	Interval score	Crisp score	Interval score	Crisp score	Interval score	Crisp score
R1 ($\wp^*$ = 0.037)	[0.116, 0.199]	0.1575	[0.150, 0.460]	0.3050	[0.072, 0.142]	0.1070	[0.072, 0.142]	0.1070	[0.170, 0.330]	0.25	[0.053, 0.094]	0.0735
R2 ($\wp^*$ = 0.035)	[0.111, 0.189]	0.1500	[0.161, 0.315]	0.2380	[0.050, 0.090]	0.0700	[0.069, 0.135]	0.1020	[0.142, 0.438]	0.2900	[0.111, 0.189]	0.1500
R3 ($\wp^*$ = 0.032)	[0.151, 0.175]	0.1630	[0.245, 0.405]	0.3250	[0.085, 0.125]	0.1050	[0.079, 0.083]	0.0810	[0.151, 0.175]	0.1630	[0.151, 0.175]	0.1630
R4 ($\wp^*$ = 0.011)	[0.036, 0.527]	0.2815	[0.023, 0.215]	0.1190	[0.023, 0.215]	0.1190	[0.019, 0.148]	0.0835	[0.039, 0.358]	0.1985	[0.039, 0.358]	0.1985
H1 ($\wp^*$ = 0.035)	[0.111, 0.189]	0.1500	[0.142, 0.438]	0.2900	[0.161, 0.315]	0.2380	[0.111, 0.189]	0.1500	[0.050, 0.090]	0.0700	[0.069, 0.135]	0.1020
H2 ($\wp^*$ = 0.011)	[0.023, 0.215]	0.1190	[0.039, 0.358]	0.1985	[0.039, 0.358]	0.1985	[0.036, 0.527]	0.2815	[0.023, 0.215]	0.1190	[0.019, 0.148]	0.0835
H3 ($\wp^*$ = 0.013)	[0.036, 0.289]	0.1625	[0.036, 0.289]	0.1625	[0.037, 0.447]	0.2420	[0.036, 0.289]	0.1625	[0.036, 0.289]	0.1625	[0.032, 0.184]	0.1080
H4 ($\wp^*$ = 0.011)	[0.023, 0.215]	0.1190	[0.036, 0.527]	0.2815	[0.039, 0.358]	0.1985	[0.039, 0.358]	0.1985	[0.019, 0.148]	0.0835	[0.023, 0.215]	0.1190
E1 ($\wp^*$ = 0.035)	[0.069, 0.135]	0.1020	[0.111, 0.189]	0.1500	[0.111, 0.189]	0.1500	[0.050, 0.090]	0.0700	[0.161, 0.315]	0.2380	[0.142, 0.438]	0.2900
E2 ($\wp^*$ = 0.010)	[0.036, 0.332]	0.1840	[0.036, 0.332]	0.1840	[0.017, 0.137]	0.0770	[0.021, 0.199]	0.1100	[0.034, 0.488]	0.2610	[0.036, 0.332]	0.1840
E3 ($\wp^*$ = 0.013)	[0.036, 0.289]	0.1625	[0.036, 0.289]	0.1625	[0.032, 0.184]	0.1080	[0.036, 0.289]	0.1625	[0.036, 0.289]	0.1625	[0.037, 0.447]	0.2420
E4 ($\wp^*$ = 0.010)	[0.034, 0.488]	0.2610	[0.036, 0.332]	0.1840	[0.036, 0.332]	0.1840	[0.036, 0.332]	0.1840	[0.017, 0.137]	0.0770	[0.021, 0.199]	0.1100
E5 ($\wp^*$ = 0.013)	[0.036, 0.289]	0.1625	[0.037, 0.447]	0.2420	[0.036, 0.289]	0.1625	[0.036, 0.289]	0.1625	[0.032, 0.184]	0.1080	[0.036, 0.289]	0.1625
S1 ($\wp^*$ = 0.011)	[0.019, 0.148]	0.0835	[0.039, 0.358]	0.1985	[0.036, 0.527]	0.2815	[0.039, 0.358]	0.1985	[0.023, 0.215]	0.1190	[0.023, 0.215]	0.1190
S2 ($\wp^*$ = 0.013)	[0.032, 0.184]	0.1080	[0.037, 0.447]	0.2420	[0.036, 0.289]	0.1625	[0.036, 0.289]	0.1625	[0.036, 0.289]	0.1625	[0.036, 0.289]	0.1625
S3 ($\wp^*$ = 0.011)	[0.036, 0.527]	0.2815	[0.023, 0.215]	0.1190	[0.039, 0.358]	0.1985	[0.039, 0.358]	0.1985	[0.019, 0.148]	0.0835	[0.023, 0.215]	0.1190
S4 ($\wp^*$ = 0.010)	[0.036, 0.332]	0.1840	[0.036, 0.332]	0.1840	[0.034, 0.488]	0.2610	[0.036, 0.332]	0.1840	[0.017, 0.137]	0.0770	[0.021, 0.199]	0.1100
C1 ($\wp^*$ = 0.011)	[0.039, 0.358]	0.1985	[0.039, 0.358]	0.1985	[0.019, 0.148]	0.0835	[0.023, 0.215]	0.1190	[0.036, 0.527]	0.2815	[0.023, 0.215]	0.1190
C2 ($\wp^*$ = 0.010)	[0.034, 0.488]	0.2610	[0.036, 0.332]	0.1840	[0.017, 0.137]	0.0770	[0.021, 0.199]	0.1100	[0.036, 0.332]	0.1840	[0.036, 0.332]	0.1840
C3 ($\wp^*$ = 0.011)	[0.039, 0.358]	0.1985	[0.039, 0.358]	0.1985	[0.019, 0.148]	0.0835	[0.023, 0.215]	0.1190	[0.036, 0.527]	0.2815	[0.023, 0.215]	0.1190
C4 ($\wp^*$ = 0.011)	[0.039, 0.358]	0.1985	[0.036, 0.527]	0.2815	[0.039, 0.358]	0.1985	[0.023, 0.215]	0.1190	[0.019, 0.148]	0.0835	[0.023, 0.215]	0.1190
C5 ($\wp^*$ = 0.010)	[0.036, 0.332]	0.1840	[0.021, 0.199]	0.1100	[0.017, 0.137]	0.0770	[0.034, 0.488]	0.2610	[0.036, 0.332]	0.1840	[0.036, 0.332]	0.1840
C6 ($\wp^*$ = 0.013)	[0.036, 0.289]	0.1625	[0.036, 0.289]	0.1625	[0.036, 0.289]	0.1625	[0.037, 0.447]	0.2420	[0.036, 0.289]	0.1625	[0.032, 0.184]	0.1080
I1 ($\wp^*$ = 0.011)	[0.039, 0.358]	0.1985	[0.036, 0.527]	0.2815	[0.023, 0.215]	0.1190	[0.039, 0.358]	0.1985	[0.019, 0.148]	0.0835	[0.023, 0.215]	0.1190
I2 ($\wp^*$ = 0.010)	[0.036, 0.332]	0.1840	[0.034, 0.488]	0.2610	[0.036, 0.332]	0.1840	[0.036, 0.332]	0.1840	[0.021, 0.199]	0.1100	[0.017, 0.137]	0.0770
I3 ($\wp^*$ = 0.013)	[0.036, 0.289]	0.1625	[0.037, 0.447]	0.2420	[0.036, 0.289]	0.1625	[0.036, 0.289]	0.1625	[0.036, 0.289]	0.1625	[0.032, 0.184]	0.1080
I4 ($\wp^*$ = 0.011)	[0.036, 0.527]	0.2815	[0.039, 0.358]	0.1985	[0.039, 0.358]	0.1985	[0.023, 0.215]	0.1190	[0.019, 0.148]	0.0835	[0.023, 0.215]	0.1190

not cause a noticeable change in the final weights of the sub-factors, which indicates the robustness of the proposed approach. On the other hand, the results of Table 18 show that the ranks of suppliers 1, 2, 3, and 5 remained constant in all scenarios with the changes made in the range of interval numbers, and only the ranks of suppliers 4 and 6 changed in two scenarios. This variation occurs because their final scores are very close, making their rankings more sensitive to minor adjustments. Therefore, it can be concluded that the proposed approach is not highly sensitive to input data and that experts' inadvertent errors do not have a significant impact on the results, which means the robustness of the proposed approach.

On the whole, four experts embarked on validating the initial set of six factors and 27 sub-factors to make sure about the accuracy and precision of the expert-based evaluation. Despite the availability of some conceptual overlap among specific sub factors such as information sharing vs. platforms, capability vs. reputation, and so on, these interdependencies are captured by the presented Supermatrix structure. This provides the framework with the possibility of computing the dependent weights reflecting the realistic interconnected nature of supplier evaluation. In terms of data collection, a brainstorming session was carried out to facilitate the procedure and the recording of all pairwise comparisons was done after achieving an agreement among the

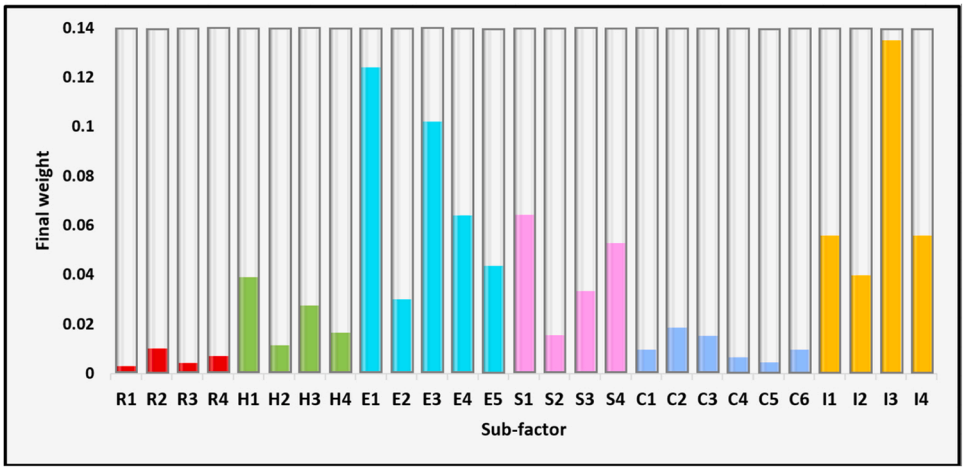


Fig. 5. The final weights of the sub-factors.

Table 17
The final weights of sub-factors in the different scenarios.

Sub-factor	Case study	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Scenario 6
R1	0.0030	0.0030	0.0032	0.0030	0.0030	0.0030	0.0032
R2	0.0101	0.0102	0.0098	0.0102	0.0101	0.0102	0.0098
R3	0.0043	0.0043	0.0044	0.0043	0.0043	0.0043	0.0044
R4	0.0072	0.0071	0.0073	0.0072	0.0072	0.0071	0.0073
H1	0.0390	0.0394	0.0376	0.0390	0.0390	0.0394	0.0376
H2	0.0115	0.0114	0.0123	0.0115	0.0115	0.0114	0.0123
H3	0.0275	0.0274	0.0279	0.0275	0.0275	0.0274	0.0279
H4	0.0166	0.0165	0.0169	0.0166	0.0166	0.0165	0.0169
E1	0.1240	0.1251	0.1222	0.1240	0.1240	0.1251	0.1222
E2	0.0300	0.0300	0.0305	0.0300	0.0300	0.0300	0.0305
E3	0.1020	0.1007	0.1033	0.1020	0.1020	0.1007	0.1033
E4	0.0640	0.0642	0.0640	0.0640	0.0640	0.0642	0.0640
E5	0.0436	0.0436	0.0436	0.0436	0.0436	0.0436	0.0436
S1	0.0643	0.0650	0.0635	0.0643	0.0643	0.0650	0.0635
S2	0.0156	0.0156	0.0160	0.0156	0.0156	0.0156	0.0160
S3	0.0333	0.0333	0.0332	0.0333	0.0333	0.0333	0.0332
S4	0.0529	0.0523	0.0535	0.0529	0.0529	0.0523	0.0535
C1	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096
C2	0.0186	0.0188	0.0184	0.0186	0.0186	0.0188	0.0184
C3	0.0153	0.0152	0.0155	0.0153	0.0153	0.0152	0.0155
C4	0.0066	0.0066	0.0066	0.0066	0.0066	0.0066	0.0066
C5	0.0045	0.0045	0.0046	0.0045	0.0045	0.0045	0.0046
C6	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096
I1	0.0559	0.0550	0.0575	0.0559	0.0559	0.0550	0.0575
I2	0.0398	0.0397	0.0421	0.0398	0.0398	0.0397	0.0421
I3	0.1350	0.1369	0.1296	0.1350	0.1350	0.1369	0.1296
I4	0.0559	0.0550	0.0575	0.0559	0.0559	0.0550	0.0575

Table 18
The score of suppliers in the different scenarios.

Supplier	Case study		Scenario 1		Scenario 2		Scenario 3		Scenario 4		Scenario 5		Scenario 6	
	Score	Rank	Score	Rank	Score	Rank	Score	Rank	Score	Rank	Score	Rank	Score	Rank
SU1	0.1727	2	0.1725	2	0.1731	2	0.1720	2	0.1720	2	0.1718	2	0.1724	2
SU2	0.2038	1	0.2038	1	0.2038	1	0.2078	1	0.1998	1	0.2079	1	0.1998	1
SU3	0.1695	3	0.1696	3	0.1694	3	0.1685	3	0.1699	3	0.1685	3	0.1698	3
SU4	0.1546	4	0.1545	4	0.1549	4	0.1538	5	0.1560	4	0.1537	5	0.1563	4
SU5	0.1454	6	0.1456	6	0.1450	6	0.1438	6	0.1477	6	0.1440	6	0.1474	6
SU6	0.1540	5	0.1541	5	0.1539	5	0.1542	4	0.1546	5	0.1541	4	0.1544	5

experts, which eliminated any possible individual subjectivity and bias. At last, a sensitivity analysis with six scenarios was conducted and it was concluded that insignificant variations in expert opinions do not result in significant changes in the final rankings, which represents the robustness of the proposed approach.

6. Discussion

In this article, a Supermatrix-based interval GBWM was presented for calculating the dependent weights of criteria and the alternatives were prioritized. Due to the fact that the proposed approach is structured based on the studied problem, some of its features are not highlighted

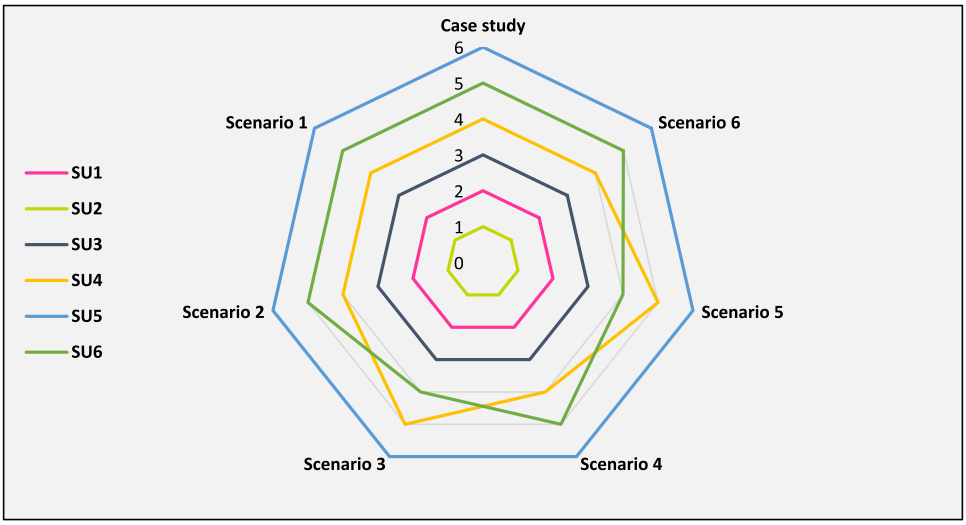


Fig. 6. The rank of suppliers in the different scenarios.

and they will be focused on in this section. In this article, the investigated network structure, as shown in Fig. 2, includes the echelons of goal, factors, sub-factors, and alternatives. However, this does not mean that this approach only supports the networks that include these four echelons. In other words, there is no limit on the number of echelons of the network. For example, if sub-factors also consist of a number of components, we should add an echelon between the sub-factors and alternatives in the network.

Another related thing is the dependency between network components. In this article, to reduce the amount of calculations, we have assumed that there is a dependency between the factors. Now the question arises that if the dependency is between the sub-factors, can this approach be used? The response to this question is positive. In Fig. 7, a comprehensive pattern is presented for structuring the initial Supermatrix, which gives a precise answer to this question.

Below is the concept of each cell of the matrix shown in Fig. 7:

- **A1**: This cell is always equal to zero.
- **A2**: The independent weights of the factors should be placed in this cell.
- **A3**: Because there is no direct relationship between the sub-factors and the goal, the values of this cell are zero.
- **A4**: Because there is no direct relationship between the alternatives and the goal, the values of this cell are zero.
- **A5**: The values of this cell are always zero.
- **A6**: If there is dependency between the factors, the dependency matrix calculated by steps 6–11 should be placed in this cell.

	Goal	Factor	Sub-factor	Alternative
Goal	**A1**	**A5**	**A9**	**A13**
Factor	**A2**	**A6**	**A10**	**A14**
Sub-factor	**A3**	**A7**	**A11**	**A15**
Alternative	**A4**	**A8**	**A12**	**A16**

Fig. 7. Comprehensive structure of initial Supermatrix.

- **A7**: The independent weights of the sub-factors should be placed in this cell.
- **A8**: Because there is no direct relationship between the alternatives and the factors, the values of this cell are zero.
- **A9**: The values of this cell are always zero.
- **A10**: Because each factor consists of a set of sub-factors, the values of this cell should be zero.
- **A11**: If there is dependency between the sub-factors, the dependency matrix calculated by steps 6–11 should be placed in this cell.
- **A12**: In this cell, the relative scores of the alternatives for each sub-factor are placed. It should be noted that these scores are calculated by step 12.
- **A13**: The values of this cell are always zero.
- **A14**: The values of this cell are always zero.
- **A15**: The values of this cell are always zero.
- **A16**: Because the alternatives are independent of each other, the values of this cell should be zero.

Although methods such as grey/interval BWM use interval numbers to calculate weights of criteria, these methods do not consider interdependencies between criteria in the weighting process. In other words, these methods calculate independent weights of criteria. Now that the proposed approach has been comprehensively explained, the question may be raised: how much does the dependency between factors (sub-factors) affect the independent weights of factors and sub-factors? To respond to this question, we should compare the independent weights of factors and sub-factors with the weights obtained from the proposed approach. The independent weights of factors are given in Table 9. To calculate the independent weights of sub-factors, we must multiply the independent weights of factors by the crisp independent weights of sub-factors. Table 19 represents the independent weights of sub-factors. On the other hand, Table B4 shows the dependent weights of factors and sub-factors. Table 19 provides a comparison between the independent and dependent weights of factors and sub-factors.

The results presented in Table 19 reveal that it is necessary to consider dependency in calculating the weights of the factors and sub-factors. For example, in sub-factor "R3", there is a 995% difference between independent and dependent weights. For a visual comparison of the results, Fig. 8 provides a schematic comparison between the independent and dependent weights of the sub-factors.

As Fig. 8 shows, when the factors are independent of each other, the sub-factors of cost (E1) and diversity and inclusion (S2) are the most and least important sub-factors, respectively. While taking into account the dependency between the factors, the sub-factors of security (I3) and

Table 19
Comparison of independent and dependent weights of factors and sub-factors.

Factor	Independent weight (ϑ_j^{CR})	Dependent weight ($\overline{\vartheta}_j^{CR}$)	$\frac{\vartheta_j^{CR}-\overline{\vartheta}_j^{CR}}{\overline{\vartheta}_j^{CR}}\times 100$	Sub-factor	Independent weight (ϑ_j^{CR})	Dependent weight ($\overline{\vartheta}_j^{CR}$)	$\frac{\vartheta_j^{CR}-\overline{\vartheta}_j^{CR}}{\overline{\vartheta}_j^{CR}}\times 100$
Resiliency	0.2500	0.025	900%	R1	0.0304	0.003	913.33%
				R2	0.1031	0.01	931%
				R3	0.0438	0.004	995%
				R4	0.0728	0.007	940%
Human-centricity	0.1575	0.095	65.79%	H1	0.065	0.039	66.67%
				H2	0.0191	0.011	73.64%
				H3	0.0458	0.028	63.57%
				H4	0.0276	0.017	62.35%
Economic	0.3050	0.364	−16.21%	E1	0.104	0.124	−16.13%
				E2	0.0252	0.03	−16%
				E3	0.0856	0.102	−16.08%
				E4	0.0537	0.064	−16.09%
				E5	0.0366	0.044	−16.82%
Social	0.0735	0.166	−55.72%	S1	0.0284	0.064	−55.63%
				S2	0.0069	0.016	−56.88%
				S3	0.0147	0.033	−55.45%
				S4	0.0234	0.053	−55.85%
Circular	0.1070	0.064	67.19%	C1	0.0161	0.01	61%
				C2	0.031	0.019	63.16%
				C3	0.0255	0.015	70%
				C4	0.0109	0.007	55.71%
				C5	0.0075	0.004	87.5%
				C6	0.0161	0.01	61%
Industry 4.0	0.1070	0.287	−62.72%	I1	0.0209	0.056	−62.68%
				I2	0.0149	0.04	−62.75%
				I3	0.0504	0.135	−62.67%
				I4	0.0209	0.056	−62.68%

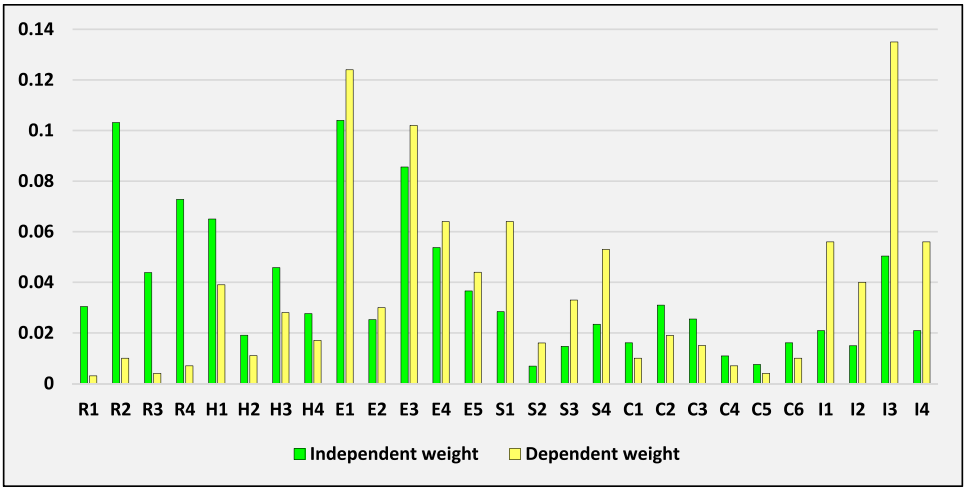


Fig. 8. Comparison of independent and dependent weights of sub-factors.

backup suppliers (R1) are the most and least important sub-factors, respectively. Therefore, it can be concluded that in MCDM problems where there is dependency between factors, the ignorance of dependency can seriously affect the weights of factors and lead to incorrect results.

7. Theoretical implications

Before the unveiling of BWM, the AHP method was regarded as one of the most popular MCDM methods among researchers. Rezaei [88] introduced the BWM and it was highly welcomed by researchers due to its high consistency and low number of pairwise comparisons. But the introduced BWM, along with its advantages, suffers from one main drawback. Due to the use of a non-linear model to calculate the weights of factors, this method sometimes leads to the production of multiple optimal solutions. To deal with this challenge, Rezaei [89] presented a linear model for calculating the weights of the factors, which solved this problem. Afterwards, numerous researchers focused on developing this method. Then, researchers put an emphasize on developing this method and improving its performance. For example, this method calculates the independent weights of factors and ignores interdependencies between factors in the weighting process. In this regard, some researchers combined this method with the methods that determine the cause-and-effect relationship among the factors to calculate the dependent weights of factors. Debnath et al. [14] combined BWM with interpretive structural modeling (ISM) to calculate the dependent weights of factors, while Tavana et al. [90] integrated the BWM and WINGS to obtain dependent weights. Similarly, to achieve this goal, some researchers calculated the interdependencies between factors by DEMATEL method and applied it to the independent weights obtained from BWM [32,91,92]. However,

Tavana et al. [31] developed a novel integrated method based on BWM to calculate the dependent weights of factors and called it GBWM. Their method first calculates the independent weights of the factors. Then, the interdependency matrix is calculated and multiplied by the independent weights of the factors. Tavana et al. [47] developed this method under uncertainty by integrating fuzzy theory and GBWM. Although GBWM is a suitable method for weighting the intertwined factors, this method does not systematically prioritize alternatives. It is noteworthy that this does not mean that GBWM is not suitable for prioritizing alternatives, but it means that this method has not provided a solution for prioritizing alternatives in an integrated way, and implements prioritization in a separate phase. In this vein, this paper, for the first time, proposed an integrated approach to calculate the dependent weights of criteria and the scores of alternatives in intertwined hierarchical networks under uncertainty by integrating GBWM, Supermatrix structure, and interval numbers.

The methodology used in this study contained the integration of GBWM, interval numbers, and a Supermatrix structure, which provides the possibility of making a more inclusive assessment and overcoming the limitations of old weighting and ranking methods. The presented approach practically benefits from robustness and adaptability. In this way, managers can go for the identification and prioritization of supplier evaluation criteria with transparency according to stakeholder expectations, their strategic goals, and rules and standards. The employment of interval numbers overcomes uncertainty and facilitates expert judgment flexibility, which makes this approach user-friendly in group decision-making contexts. In addition, the integrated networked analyses are supported by the Supermatrix-based structure, which is required for dealing with complex supplier relationships in large construction projects.

8. Managerial implications

The concept of circular supplier 5.0 selection is presented and operationalized in the current study. This provides insightful tips for the decision-makers involved in this area by upgrading the theoretical and practical foundations of supplier selection in the construction industry. When CE principles and Industry 5.0 dimensions are systematically integrated with the supplier evaluation procedure, the progressive vision of sustainability and resilience in the construction area is enhanced.

The findings of this study can distinguish a number of important hints pertaining to management. From among the 27 evaluated factors, the top three ones, namely security, cost, and quality are recognized as the most determining ones in circular supplier 5.0 selection. The categorization of criteria in this way shows the importance of industry concentration on data and operational security. Accordingly, the prominent role of digitalization and integration assumes higher significance and also the cost and quality factors contribute to the maintenance of solid competitiveness and value delivery. In this regard, security is prior to other criteria and can lead to enhanced reliance in interconnected platforms and digital supply chains. This will expose companies to new operational and cybersecurity risks. Thus, it is necessary for managers to assign credit to solid security protocols and demand evidence of digital resilience. In the same way, cost and quality should be given priority in making procurement decisions. It is noteworthy that, in the realm of the circular supplier 5.0 context, these criteria are assessed in accordance with sustainability and innovation.

According to the results, information sharing and traceability are the factors that are considered the most important sub-factors. This leads to a higher level of transparency and collaboration in the supplier chain in consistence with the human-centric and interconnected ethos of Industry 5.0. The superiority of these sub-criteria recommends that construction companies should pay more attention to digital infrastructure and provide open communication channels within the supply networks to bring strong compliance, trust, and adaptability.

The comparison of the current results with the previous ones shows both convergence and divergence. The significance of sustainability and economic criteria has been emphasized in prior studies (e.g., [22,23]); however, only a limited number of studies have embarked on integrating circularity and Industry 5.0 in a holistic approach. The current findings are in line with the findings reported by Bag and Kumar Mangla [26] and Mishra et al. [25] in terms of circular criteria while these findings go further in proving the relatively equal importance of digital security and information-based features with traditional economic factors.

Similar to other expert-based evaluation tasks, there is the risk of bias in obtaining expert opinion. Such subjective features as organizational culture, individual experiences, and limited exposure to emerging trends are likely to contaminate expert judgment. In order to minimize this bias, the present study made use of a variety of experts from different professional backgrounds and levels and also utilized interval numbers to obtain a wide range of opinions. The managers using this method are recommended to make sure about diversity among decision-makers so that they can give sufficient information on the evaluation factors and complement expert opinion by means of numerical and historical performance data whenever possible.

Another benefit of this approach is its scalability. As it was shown in the construction context, this method has a modular design and capacity in dealing with a large number of criteria and alternatives. This feature makes it employable in other industries that encounter such critical issues. Besides, the companies with more comprehensive datasets or more complex supply networks can securely employ this approach.

On the whole, this study enriches the theoretical foundation of this area as it introduces a novel MCDM framework and the circular supplier 5.0 concept. Practically, it is also highly useful as it provides the managers seeking to assess their supplier selection strategies with a sound practical tool. If companies put a holistic set of criteria into practice, they can boost the level of competitiveness, guarantee the observance of rules and standards, and assist with the realization of sustainability goals. The reliability and practical value of this method is approved by the positive feedback received from industry experts; therefore, its applicability in the present and future supplier selection projects is strongly recommended.

9. Conclusion

In transition from linear economy to CE, Industry 4.0 acts as a catalyst, moving the supply chain towards sustainability more quickly. However, because Industry 4.0 prioritizes machines over humans, it conflicts with some sustainability goals. Applying Industry 5.0 can overcome this challenge since it emphasizes the integration of human values with advanced technologies. In this vein, this article introduced for the first time a new concept called circular supplier 5.0 for evaluating suppliers in the construction industry by integrating the criteria of Industry 5.0 (i.e., sustainability, human-centricity, and resiliency), Industry 4.0, and circularity. Furthermore, by integrating GBWM, Supermatrix structure, and interval numbers, a holistic approach was developed to calculate the weights of factors and sub-factors and rank suppliers in a hierarchical intertwined network. The effectiveness of the proposed approach was examined by implementing it in a Malaysian construction company where the performance of six suppliers of this company was evaluated using six factors (i.e., resiliency, human-centricity, economic, circular, social, and Industry 4.0) along with their 27 sub-factors. The results showed that modern sub-factors such as security, information sharing, and traceability were as important as traditional sub-factors such as cost, quality, and capability. Then, the robustness of the proposed approach was examined using a sensitivity analysis process. Finally, by implementing a comparative analysis, it was shown that the interdependency item had a significant impact on the weights of the factors and sub-factors, and this item should not be ignored in intertwined networks.

Although this article makes a great practical and theoretical contribution to the circular supplier 5.0 selection problem, it comes with some

limitations that are suggested to future researchers to focus on. The results of the sensitivity analysis process showed that the use of interval numbers and consensus-based brainstorming method reduced individual biases. However, due to practical limitations, the developed approach was validated only based on the opinions of four experts from a construction company, which may raise concerns about the generalizability of the findings. Therefore, it is suggested that a larger panel of experts be applied in future research to reduce biases and enable generalizability of the findings. Furthermore, this article presents a subjective approach to evaluating circular supplier 5.0 that relies on expert judgments. To reduce the subjectivity of the proposed approach, it is suggested that objective (data-driven) approaches be combined with expert opinions to establish a logical balance between objective and subjective inputs. In addition, the proposed approach has been used to prioritize six suppliers in the construction sector. Given that the proposed approach has no limits on the number of alternatives and is not exclusive to the construction industry, it is suggested that proposed approach be applied in other sectors such as automotive, food industries, etc. that have more suppliers.

CRediT authorship contribution statement

Esmaeel Saberi: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Arash Khalili Nasr:** Writing – review & editing, Writing – original draft, Conceptualization. **Hassan Mina:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shahryar Sorooshian:** Writing – review & editing, Supervision, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The author used ChatGPT (version 5.3) to improve the presentation of information. All content was reviewed and verified by the author.

Appendix A

Table A1
The best-to-others vector for resiliency sub-factors

	R1	R2	R3	R4
Best: R2	[5/2, 7/2]	[1, 1]	[3/2, 5/2]	[2/3, 3/2]

Table A2
The best-to-others vector for human-centricity sub-factors

	H1	H2	H3	H4
Best: H1	[1, 1]	[5/2, 7/2]	[2/3, 3/2]	[3/2, 5/2]

Table A3
The best-to-others vector for economic sub-factors

	E1	E2	E3	E4	E5
Best: E1	[1, 1]	[7/2, 9/2]	[2/3, 3/2]	[3/2, 5/2]	[5/2, 7/2]

Table A4
The best-to-others vector for social sub-factors

	S1	S2	S3	S4
Best: S1	[1, 1]	[7/2, 9/2]	[3/2, 5/2]	[2/3, 3/2]

Table A5
The best-to-others vector for circular sub-factors

	C1	C2	C3	C4	C5	C6
Best: C2	[3/2, 5/2]	[1, 1]	[2/3, 3/2]	[5/2, 7/2]	[7/2, 9/2]	[3/2, 5/2]

Table A6
The best-to-others vector for Industry 4.0 sub-factors

	I1	I2	I3	I4
Best: I3	[3/2, 5/2]	[5/2, 7/2]	[1, 1]	[3/2, 5/2]

Table A7
The others-to-worst vector for resiliency sub-factor

	Worst: R1
R1	[1, 1]
R2	[5/2, 7/2]
R3	[2/3, 3/2]
R4	[3/2, 5/2]

Table A8
The others-to-worst vector for human-centricity sub-factor

	Worst: H2
H1	[5/2, 7/2]
H2	[1, 1]
H3	[3/2, 5/2]
H4	[2/3, 3/2]

Table A9
The others-to-worst vector for economic sub-factor

	Worst: E2
E1	[7/2, 9/2]
E2	[1, 1]
E3	[5/2, 7/2]
E4	[3/2, 5/2]
E5	[2/3, 3/2]

Table A10
The others-to-worst vector for social sub-factor

	Worst: S2
S1	[7/2, 9/2]
S2	[1, 1]
S3	[3/2, 5/2]
S4	[5/2, 7/2]

Table A11
The others-to-worst vector for circular sub-factor

	Worst: C5
C1	[3/2, 5/2]
C2	[7/2, 9/2]
C3	[5/2, 7/2]
C4	[2/3, 3/2]
C5	[1, 1]
C6	[3/2, 5/2]

Table A12
The others-to-worst vector for Industry 4.0 sub-factor

	Worst: I2
I1	[2/3, 3/2]
I2	[1, 1]
I3	[5/2, 7/2]
I4	[2/3, 3/2]

Appendix B

Table B1
The best-to-others vector for evaluating suppliers for each sub-factor

R1	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU2	[3/2, 5/2]	[1, 1]	[5/2, 7/2]	[5/2, 7/2]	[2/3, 3/2]	[7/2, 9/2]
R2	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU5	[3/2, 5/2]	[2/3, 3/2]	[7/2, 9/2]	[5/2, 7/2]	[1, 1]	[3/2, 5/2]
R3	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU2	[3/2, 5/2]	[1, 1]	[5/2, 7/2]	[7/2, 9/2]	[3/2, 5/2]	[3/2, 5/2]
R4	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU1	[1, 1]	[3/2, 5/2]	[3/2, 5/2]	[5/2, 7/2]	[2/3, 3/2]	[2/3, 3/2]
H1	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU2	[3/2, 5/2]	[1, 1]	[2/3, 3/2]	[3/2, 5/2]	[7/2, 9/2]	[5/2, 7/2]
H2	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU4	[3/2, 5/2]	[2/3, 3/2]	[2/3, 3/2]	[1, 1]	[3/2, 5/2]	[5/2, 7/2]
H3	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU3	[2/3, 3/2]	[2/3, 3/2]	[1, 1]	[2/3, 3/2]	[2/3, 3/2]	[3/2, 5/2]
H4	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU2	[3/2, 5/2]	[1, 1]	[2/3, 3/2]	[2/3, 3/2]	[5/2, 7/2]	[3/2, 5/2]
E1	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU6	[5/2, 7/2]	[3/2, 5/2]	[3/2, 5/2]	[7/2, 9/2]	[2/3, 3/2]	[1, 1]
E2	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU5	[2/3, 3/2]	[2/3, 3/2]	[5/2, 7/2]	[3/2, 5/2]	[1, 1]	[2/3, 3/2]
E3	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU6	[2/3, 3/2]	[2/3, 3/2]	[3/2, 5/2]	[2/3, 3/2]	[2/3, 3/2]	[1, 1]
E4	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU1	[1, 1]	[2/3, 3/2]	[2/3, 3/2]	[2/3, 3/2]	[5/2, 7/2]	[3/2, 5/2]
E5	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU2	[2/3, 3/2]	[1, 1]	[2/3, 3/2]	[2/3, 3/2]	[3/2, 5/2]	[2/3, 3/2]
S1	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU3	[5/2, 7/2]	[2/3, 3/2]	[1, 1]	[2/3, 3/2]	[3/2, 5/2]	[3/2, 5/2]
S2	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU2	[3/2, 5/2]	[1, 1]	[2/3, 3/2]	[2/3, 3/2]	[2/3, 3/2]	[2/3, 3/2]
S3	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU1	[1, 1]	[3/2, 5/2]	[2/3, 3/2]	[2/3, 3/2]	[5/2, 7/2]	[3/2, 5/2]
S4	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU3	[2/3, 3/2]	[2/3, 3/2]	[1, 1]	[2/3, 3/2]	[5/2, 7/2]	[3/2, 5/2]
C1	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU5	[2/3, 3/2]	[2/3, 3/2]	[5/2, 7/2]	[3/2, 5/2]	[1, 1]	[3/2, 5/2]
C2	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU1	[1, 1]	[2/3, 3/2]	[5/2, 7/2]	[3/2, 5/2]	[2/3, 3/2]	[2/3, 3/2]
C3	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU5	[2/3, 3/2]	[2/3, 3/2]	[5/2, 7/2]	[3/2, 5/2]	[1, 1]	[3/2, 5/2]
C4	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**

(*continued on next page*)

Table B1 (*continued*)

R1	SU1	SU2	SU3	SU4	SU5	SU6
Best: SU2	[2/3, 3/2]	[1, 1]	[2/3, 3/2]	[3/2, 5/2]	[5/2, 7/2]	[3/2, 5/2]
C5	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU4	[2/3, 3/2]	[3/2, 5/2]	[5/2, 7/2]	[1, 1]	[2/3, 3/2]	[2/3, 3/2]
C6	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU4	[2/3, 3/2]	[2/3, 3/2]	[2/3, 3/2]	[1, 1]	[2/3, 3/2]	[3/2, 5/2]
I1	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU2	[2/3, 3/2]	[1, 1]	[3/2, 5/2]	[2/3, 3/2]	[5/2, 7/2]	[3/2, 5/2]
I2	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU2	[2/3, 3/2]	[1, 1]	[2/3, 3/2]	[2/3, 3/2]	[3/2, 5/2]	[5/2, 7/2]
I3	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU2	[2/3, 3/2]	[1, 1]	[2/3, 3/2]	[2/3, 3/2]	[2/3, 3/2]	[3/2, 5/2]
I4	**SU1**	**SU2**	**SU3**	**SU4**	**SU5**	**SU6**
Best: SU1	[1, 1]	[2/3, 3/2]	[2/3, 3/2]	[3/2, 5/2]	[5/2, 7/2]	[3/2, 5/2]

Table B2
The others-to-worst vector for evaluating suppliers for each sub-factor

R1	Worst: SU6	R2	Worst: SU3	R3	Worst: SU4	R4	Worst: SU4	H1	Worst: SU5	H2	Worst: SU6	H3	Worst: SU6	H4	Worst: SU5	E1	Worst: SU4
SU1	[3/2, 5/2]	SU1	[3/2, 5/2]	SU1	[3/2, 5/2]	SU1	[5/2, 7/2]	SU1	[3/2, 5/2]	SU1	[2/3, 3/2]	SU1	[2/3, 3/2]	SU1	[2/3, 3/2]	SU1	[2/3, 3/2]
SU2	[7/2, 9/2]	SU2	[5/2, 7/2]	SU2	[7/2, 9/2]	SU2	[2/3, 3/2]	SU2	[7/2, 9/2]	SU2	[3/2, 5/2]	SU2	[2/3, 3/2]	SU2	[5/2, 7/2]	SU2	[3/2, 5/2]
SU3	[2/3, 3/2]	SU3	[1, 1]	SU3	[2/3, 3/2]	SU3	[2/3, 3/2]	SU3	[5/2, 7/2]	SU3	[3/2, 5/2]	SU3	[3/2, 5/2]	SU3	[3/2, 5/2]	SU3	[3/2, 5/2]
SU4	[2/3, 3/2]	SU4	[2/3, 3/2]	SU4	[1, 1]	SU4	[1,1]	SU4	[3/2, 5/2]	SU4	[5/2, 7/2]	SU4	[2/3, 3/2]	SU4	[3/2, 5/2]	SU4	[1, 1]
SU5	[5/2, 7/2]	SU5	[7/2, 9/2]	SU5	[3/2, 5/2]	SU5	[3/2, 5/2]	SU5	[1, 1]	SU5	[2/3, 3/2]	SU5	[2/3, 3/2]	SU5	[1, 1]	SU5	[5/2, 7/2]
SU6	[1, 1]	SU6	[3/2, 5/2]	SU6	[3/2, 5/2]	SU6	[3/2, 5/2]	SU6	[2/3, 3/2]	SU6	[1, 1]	SU6	[1, 1]	SU6	[2/3, 3/2]	SU6	[7/2, 9/2]
E2	Worst: SU3	**E3**	Worst: SU3	**E4**	Worst: SU5	**E5**	Worst: SU5	**S1**	Worst: SU1	**S2**	Worst: SU1	**S3**	Worst: SU5	**S4**	Worst: SU5	**C1**	Worst: SU3
SU1	[3/2, 5/2]	SU1	[2/3, 3/2]	SU1	[5/2, 7/2]	SU1	[2/3, 3/2]	SU1	[1, 1]	SU1	[1, 1]	SU1	[5/2, 7/2]	SU1	[3/2, 5/2]	SU1	[3/2, 5/2]
SU2	[3/2, 5/2]	SU2	[2/3, 3/2]	SU2	[3/2, 5/2]	SU2	[3/2, 5/2]	SU2	[3/2, 5/2]	SU2	[3/2, 5/2]	SU2	[2/3, 3/2]	SU2	[3/2, 5/2]	SU2	[3/2, 5/2]
SU3	[1, 1]	SU3	[1, 1]	SU3	[3/2, 5/2]	SU3	[2/3, 3/2]	SU3	[5/2, 7/2]	SU3	[2/3, 3/2]	SU3	[3/2, 5/2]	SU3	[5/2, 7/2]	SU3	[1, 1]
SU4	[2/3, 3/2]	SU4	[2/3, 3/2]	SU4	[3/2, 5/2]	SU4	[2/3, 3/2]	SU4	[3/2, 5/2]	SU4	[2/3, 3/2]	SU4	[3/2, 5/2]	SU4	[3/2, 5/2]	SU4	[2/3, 3/2]
SU5	[5/2, 7/2]	SU5	[2/3, 3/2]	SU5	[1, 1]	SU5	[1, 1]	SU5	[2/3, 3/2]	SU5	[2/3, 3/2]	SU5	[1, 1]	SU5	[1, 1]	SU5	[5/2, 7/2]
SU6	[3/2, 5/2]	SU6	[3/2, 5/2]	SU6	[2/3, 3/2]	SU6	[2/3, 3/2]	SU6	[2/3, 3/2]	SU6	[2/3, 3/2]	SU6	[2/3, 3/2]	SU6	[2/3, 3/2]	SU6	[2/3, 3/2]
C2	Worst: SU3	**C3**	Worst: SU3	**C4**	Worst: SU5	**C5**	Worst: SU3	**C6**	Worst: SU6	**I1**	Worst: SU5	**I2**	Worst: SU6	**I3**	Worst: SU6	**I4**	Worst: SU5
SU1	[5/2, 7/2]	SU1	[3/2, 5/2]	SU1	[3/2, 5/2]	SU1	[3/2, 5/2]	SU1	[2/3, 3/2]	SU1	[3/2, 5/2]	SU1	[3/2, 5/2]	SU1	[2/3, 3/2]	SU1	[5/2, 7/2]
SU2	[3/2, 5/2]	SU2	[3/2, 5/2]	SU2	[5/2, 7/2]	SU2	[2/3, 3/2]	SU2	[2/3, 3/2]	SU2	[5/2, 7/2]	SU2	[5/2, 7/2]	SU2	[3/2, 5/2]	SU2	[3/2, 5/2]
SU3	[1, 1]	SU3	[1, 1]	SU3	[3/2, 5/2]	SU3	[1, 1]	SU3	[2/3, 3/2]	SU3	[2/3, 3/2]	SU3	[3/2, 5/2]	SU3	[2/3, 3/2]	SU3	[3/2, 5/2]
SU4	[2/3, 3/2]	SU4	[2/3, 3/2]	SU4	[2/3, 3/2]	SU4	[5/2, 7/2]	SU4	[3/2, 5/2]	SU4	[3/2, 5/2]	SU4	[3/2, 5/2]	SU4	[2/3, 3/2]	SU4	[2/3, 3/2]
SU5	[3/2, 5/2]	SU5	[5/2, 7/2]	SU5	[1, 1]	SU5	[3/2, 5/2]	SU5	[2/3, 3/2]	SU5	[1, 1]	SU5	[2/3, 3/2]	SU5	[2/3, 3/2]	SU5	[1, 1]
SU6	[3/2, 5/2]	SU6	[2/3, 3/2]	SU6	[2/3, 3/2]	SU6	[3/2, 5/2]	SU6	[1, 1]	SU6	[2/3, 3/2]	SU6	[1, 1]	SU6	[1, 1]	SU6	[2/3, 3/2]

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Table B3
Initial Supermatrix

	Goal	R	H	E	S	C	I	R1	R2	R3	R4	H1	H2	H3	H4	E1	E2	E3	E4	E5	S1	S2	S3	S4	C1	C2	C3	C4	C5	C6	I1	I2	I3	I4	SU1	SU2	SU3	SU4	SU5	SU6
Goal	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R	0.25	0.5000	0.13	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
H	0.1575	0.1185	0.5000	0.0740	0.1052	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	0.3050	0.2130	0.13	0.5000	0	0.3260	0.5000	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S	0.0735	0.0500	0.13	0.1760	0.5000	0.0870	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C	0.1070	0	0.0550	0.0740	0	0.5000	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
I	0.1070	0.1185	0.0550	0.1760	0.3948	0.0870	0.5000	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R1	0	0.1215	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R2	0	0.4125	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R3	0	0.175	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R4	0	0.2910	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
H1	0	0	0.4125	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
H2	0	0	0.1215	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
H3	0	0	0.2910	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
H4	0	0	0.175	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E1	0	0	0	0.341	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E2	0	0	0	0.0825	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E3	0	0	0	0.2805	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E4	0	0	0	0.1760	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E5	0	0	0	0.1200	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S1	0	0	0	0	0.387	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S2	0	0	0	0	0.0940	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S3	0	0	0	0	0.2005	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S4	0	0	0	0	0.3185	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C1	0	0	0	0	0	0.1500	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C2	0	0	0	0	0	0.2900	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C3	0	0	0	0	0	0.2380	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C4	0	0	0	0	0	0.1020	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C5	0	0	0	0	0	0.07	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C6	0	0	0	0	0	0.1500	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
I1	0	0	0	0	0	0	0.1950	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
I2	0	0	0	0	0	0	0.1390	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
I3	0	0	0	0	0	0	0.471	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
I4	0	0	0	0	0	0	0.1950	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SU1	0	0	0	0	0	0	0	0.1575	0.15	0.163	0.2815	0.15	0.119	0.1625	0.119	0.102	0.184	0.1625	0.261	0.1625	0.0835	0.108	0.2815	0.184	0.1985	0.261	0.1985	0.1985	0.184	0.1625	0.1985	0.184	0.1625	0.2815	0	0	0	0	0
SU2	0	0	0	0	0	0	0	0.305	0.238	0.325	0.119	0.29	0.1985	0.1625	0.2815	0.15	0.184	0.1625	0.184	0.242	0.1985	0.242	0.119	0.184	0.1985	0.184	0.1985	0.2815	0.11	0.1625	0.2815	0.261	0.242	0.1985	0	0	0	0	0
SU3	0	0	0	0	0	0	0	0.107	0.07	0.105	0.119	0.238	0.1985	0.242	0.1985	0.15	0.077	0.108	0.184	0.1625	0.2815	0.1625	0.1985	0.261	0.0835	0.077	0.0835	0.1985	0.077	0.1625	0.119	0.184	0.1625	0.1985	0	0	0	0	0
SU4	0	0	0	0	0	0	0	0.107	0.102	0.081	0.0835	0.15	0.2815	0.1625	0.1985	0.07	0.11	0.1625	0.184	0.1625	0.1985	0.1625	0.1985	0.184	0.119	0.11	0.119	0.119	0.261	0.242	0.1985	0.184	0.1625	0.119	0	0	0	0	0
SU5	0	0	0	0	0	0	0	0.25	0.29	0.163	0.1985	0.07	0.119	0.1625	0.0835	0.238	0.261	0.1625	0.077	0.108	0.119	0.1625	0.0835	0.077	0.2815	0.184	0.2815	0.0835	0.184	0.1625	0.0835	0.11	0.1625	0.0835	0	0	0	0	0
SU6	0	0	0	0	0	0	0	0.0735	0.15	0.163	0.1985	0.102	0.0835	0.108	0.119	0.29	0.184	0.242	0.11	0.1625	0.119	0.1625	0.119	0.11	0.119	0.184	0.119	0.119	0.184	0.108	0.119	0.077	0.108	0.119	0	0	0	0	0

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Table B4
Final Supermatrix

	Goal	R	H	E	S	C	I	R1	R2	R3	R4	H1	H2	H3	H4	E1	E2	E3	E4	E5	S1	S2	S3	S4	C1	C2	C3	C4	C5	C6	I1	I2	I3	I4	SU1	SU2	SU3	SU4	SU5	SU6
Goal	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R	0.0246	0.0246	0.0246	0.0246	0.0246	0.0246	0.0246	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
H	0.0946	0.0946	0.0946	0.0946	0.0946	0.0946	0.0946	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	0.3636	0.3636	0.3636	0.3636	0.3636	0.3636	0.3636	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S	0.1662	0.1662	0.1662	0.1662	0.1662	0.1662	0.1662	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C	0.0642	0.0642	0.0642	0.0642	0.0642	0.0642	0.0642	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
I	0.2867	0.2867	0.2867	0.2867	0.2867	0.2867	0.2867	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R1	0.0030	0.0030	0.0030	0.0030	0.0030	0.0030	0.0030	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R2	0.0101	0.0101	0.0101	0.0101	0.0101	0.0101	0.0101	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R3	0.0043	0.0043	0.0043	0.0043	0.0043	0.0043	0.0043	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R4	0.0072	0.0072	0.0072	0.0072	0.0072	0.0072	0.0072	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
H1	0.0390	0.0390	0.0390	0.0390	0.0390	0.0390	0.0390	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
H2	0.0115	0.0115	0.0115	0.0115	0.0115	0.0115	0.0115	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
H3	0.0275	0.0275	0.0275	0.0275	0.0275	0.0275	0.0275	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
H4	0.0166	0.0166	0.0166	0.0166	0.0166	0.0166	0.0166	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E1	0.1240	0.1240	0.1240	0.1240	0.1240	0.1240	0.1240	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E2	0.0300	0.0300	0.0300	0.0300	0.0300	0.0300	0.0300	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E3	0.1020	0.1020	0.1020	0.1020	0.1020	0.1020	0.1020	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E4	0.0640	0.0640	0.0640	0.0640	0.0640	0.0640	0.0640	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E5	0.0436	0.0436	0.0436	0.0436	0.0436	0.0436	0.0436	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S1	0.0643	0.0643	0.0643	0.0643	0.0643	0.0643	0.0643	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S2	0.0156	0.0156	0.0156	0.0156	0.0156	0.0156	0.0156	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S3	0.0333	0.0333	0.0333	0.0333	0.0333	0.0333	0.0333	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S4	0.0529	0.0529	0.0529	0.0529	0.0529	0.0529	0.0529	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C1	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C2	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0.0186	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C3	0.0153	0.0153	0.0153	0.0153	0.0153	0.0153	0.0153	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C4	0.0066	0.0066	0.0066	0.0066	0.0066	0.0066	0.0066	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C5	0.0045	0.0045	0.0045	0.0045	0.0045	0.0045	0.0045	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
C6	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
I1	0.0559	0.0559	0.0559	0.0559	0.0559	0.0559	0.0559	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
I2	0.0398	0.0398	0.0398	0.0398	0.0398	0.0398	0.0398	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
I3	0.1350	0.1350	0.1350	0.1350	0.1350	0.1350	0.1350	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
I4	0.0559	0.0559	0.0559	0.0559	0.0559	0.0559	0.0559	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SU1	0.1727	0.1727	0.1727	0.1727	0.1727	0.1727	0.1727	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SU2	0.2038	0.2038	0.2038	0.2038	0.2038	0.2038	0.2038	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SU3	0.1695	0.1695	0.1695	0.1695	0.1695	0.1695	0.1695	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SU4	0.1546	0.1546	0.1546	0.1546	0.1546	0.1546	0.1546	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SU5	0.1454	0.1454	0.1454	0.1454	0.1454	0.1454	0.1454	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SU6	0.1540	0.1540	0.1540	0.1540	0.1540	0.1540	0.1540	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

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